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Stability of Time-Varying Receding Horizon Games

Master's Thesis

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February, 4, 2025

Abstract

This work addresses stability and recursive feasibility guarantees for receding horizon games under a potential game structure with time-varying linear quadratic objective and linear time-varying dynamics. By leveraging connections between receding horizon games and economic model predictive control theory, we derive two approaches providing asymptotic stability and recursive feasibility, once without terminal ingredients, relying on controllability assumptions, and once using an exact terminal constraint based on a dissipative periodic reference trajectory.

The proposed methods are evaluated through simulations on a demand-side management example, demonstrating their ability to handle state and input constraints while achieving nearly identical closed-loop performance. Furthermore, both approaches successfully handled positive semi-definite input costs, expanding their practical applicability.

Future research could explore extensions to nonlinear systems, non-periodic dynamics, and non-potential games, as well as relaxing assumptions for the method with terminal ingredients.

Keywords: Game-theoretic MPC, Economic MPC, Dissipativity, Turnpike.

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Nomenclature

Acronyms

w-PG	Weighted potential game
DSM	Demand-side management
EMPC	Economic model predictive control
GNE	Generalized Nash equilibrium
GNEP	Generalized Nash equilibrium problem
KKT	Karush-Kuhn-Tucker
LMI	Linear matrix inequality
LTI	Linear time-invariant
LTV	Linear time-varying
MPC	Model predictive control
NE	Nash equilibrium
NEP	Nash equilibrium problem
OCP	Optimal control problem
PG	Potential game
RHG	Receding horizon game
RK4	Runge-Kutta-4
SoC	State of charge

Chapter 1

Introduction

We live in a society driven by self-interest, governed by complex dynamics and constrained by finite resources. Numerous systems, such as traffic networks and energy grids, demonstrate how these dynamics can lead to instability, reflected in issues like traffic congestion or blackouts. While these are just two examples, similar challenges arise across a wide range of sociotechnical system in which humans are involved as decision makers. Efforts to stabilize such systems, such as traffic lights or energy storage systems, exist, but there are few theoretical results underpinning their stability.

This is where game theory comes in. As a branch of mathematics, it provides tools to model systems of self-interested agents whose decisions collectively shape the system's behavior [1]. Take traffic congestion as an example: each driver (agent) makes decisions about their route and timing to minimize their own travel time and costs, creating a competitive game [2]. These decisions are constrained by factors such as road capacity, traffic conditions, and other external influences. Game theory helps us explore how these individual motivations and constraints affect the overall system and gain insights into its stability and potential vulnerabilities.

We now aim to approximate the optimal solution to these games, that one would obtain by solving the game over an infinite time, in a receding horizon fashion. This methodology is also referred to as receding horizon games (RHGs) and can be understood as a combination of game theory and model predictive control (MPC). MPC is an optimization-based control method that dynamically adjusts the system's behavior using a mathematical model to predict the future [3]. Recently this combination has enjoyed an increasing amount of attention, as it allows to respect system constraints while achieving near-optimal performance with theoretical stability guarantees [4, 5, 6, 7, 8]. Figure 1.1 illustrates how RHGs connect to economic model predictive control (EMPC) theory.

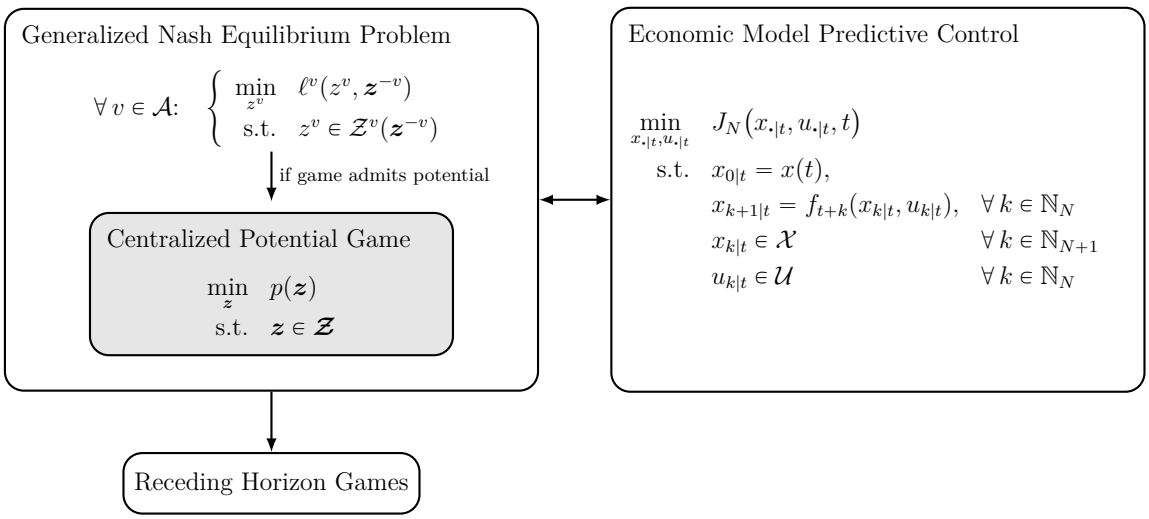

Figure 1.1: Connecting Receding Horizon Games and EMPC

1.1 Related Work

Game-theoretic control schemes have gained popularity in recent years for their potential to regulate sociotechnical systems involving self-interested agents. Unlike classical control methods, which impose hard control actions, such as traffic lights, these schemes enforce soft policies, making them more cost-effective by reducing infrastructure requirements. Receding horizon game present an especially promising control methodology due to their ability to operate such systems respecting shared resources and constraints while incorporating a dynamic model to predict near-optimal future control actions. However, formulating such control schemes is nontrivial, leading to the development of various different game-theoretic approaches.

For example [9] addresses the problem of controlling noncooperative agents, with convex cost and coupling constraints, controlled by a central coordinator with application to network congestion and demand-side management. Their approach offers a model-free dynamic control law that guarantees global convergence to a Nash equilibrium, with scalability that is independent of the population size. In [10], highway traffic control is formulated as a mixed-integer potential game with the goal of alleviating traffic congestion by incentivizing plug-in-hybrid and electric vehicles to stop at charging stations in order to reduce congestion during peak demand hours. They are able to demonstrate convergence to a Nash equilibrium, i.e., an approximate optimal charging strategy of electric vehicles, using a dynamic pricing policy, effectively showing its potential to alleviate congestion. The work of [11] models competing drivers in traffic (they refer to it as urban driving game) through lexicographic objectives together with coupling collision avoidance constraints. Under mild assumptions, they show that the game is a lexicographic potential game which enables them to establish an iterative best-response algorithm that solves the generalized Nash equilibrium problem by individually solving each driver's optimal control problem. While their simulations indicate convergence, they do not provide theoretical stability guarantees. The authors of [8] model the interaction between vehicles and their environment using a generalized potential game with an additional coupling introduced through shared collision avoidance constraints. The problem is then solved in a receding horizon fashion with a learning-based MPC scheme. In simulation they are able to show that the proposed framework is able to solve the *frozen robot problem* [12] without giving any theoretical results on stability or robustness. In [6], the authors overcome the computational burden of solving a dynamic game in a decentralized fashion by reformulating it as a centralized generalized Nash equilibrium problem under a potential game structure. This transformation allows them to solve the interactive game with coupling constraints using constrained optimization (RHG), achieving an average performance increase of a factor of 20 over leading-edge decentralized algorithms. However, their method lacks stability and recursive feasibility guarantees. Similarly [5] propose a RHG scheme to solve a stochastic traffic routing problems under a potential game structure. They are able to analytically demonstrate asymptotic convergence and provide closed-loop performance guarantees which they confirmed through numerical simulations. They further propose a semi-decentralized algorithm to solve the routing problem for nonlinear road latency functions using a nonlinear variable transformation. The work of [7] models the problem of demand-side management using an aggregative potential games (PGs), allowing it to be solved in a RHG fashion. They establish recursive feasibility and asymptotic convergence in this time-invariant setup by leveraging economic model predictive control theory, assuming convexity of both the objective and constraints. Recently these stability results have been extended to non-potential games in [4] based on dissipativity and monotone operator theory.

1.2 Contribution

Previous work has analyzed theoretical properties of Receding Horizon Games for time-invariant game-theoretic models [4, 7]. In these models, the system's dynamics and parameters remain constant over time. However, this assumption often does not hold for real-world systems, which are inherently time-varying. Consider, for example, how traffic conditions fluctuate depending on the time of year or even the time of day, influenced by factors like snow or road construction.

In this work we derive closed-loop stability guarantees for time-varying RHGs where the optimal operating behavior is given by a time-varying trajectory rather than a steady-state. To achieve this we rely on a connection between RHGs and MPC theory, first established in [7], that allows us to derive practical asymptotic stability of the optimal trajectory building on recent results from EMPC [13, 14]. In order to make this connection, we rely on a specific class of games known as potential game, where all agents unknowingly optimize a shared objective, called the potential function, which represents a common cost or goal. This unique property allows us to reframe the problem such that we can leverage MPC theory to derive closed-loop stability certificates.

The main contribution of this work are closed-loop stability guarantees for RHGs with an linear time-varying dynamics and a time-varying linear-quadratic stage cost. Hereby we demonstrate asymptotic convergence to the optimal trajectory using two different approaches. In Section 4.3 we establish stability without terminal ingredients for a setting with only input constraints, given some additional controllability assumptions, whereas in Section 4.4 we show stability with terminal ingredients for a periodically time-varying system together with the assumption of a continuous optimal rotated value function. In Chapter 5, we compare the two approaches derived in Chapter 4 through simulations.

Chapter 2

Notation

We denote the set of all integers with $\mathbb{N}$ and the real numbers by $\mathbb{R}$ respectively. Further, the set of integers in the interval $[0, N-1]$ is denoted as $\mathbb{N}_N := \{0, \dots, N-1\}$. The zero matrix and zero vector are referred to as $0_{p,q} \in \mathbb{R}^{p,q}$ and $0_m \in \mathbb{R}^m$, respectively. Similarly, the identity matrix and square identity matrix are denoted as $\mathcal{I}_{p,q} \in \mathbb{R}^{p,q}$ and $\mathcal{I}_m \in \mathbb{R}^{m,m}$. Matrices / matrix functions are generally denoted using capital letters, e.g., $A(t)$, whereas vectors / vector-valued functions are denoted using lowercase letters, such as $x(t)$. $\text{blkdiag}(A, \dots, Z)$ is used for block diagonal matrices with the block matrices A to Z on their diagonal. Furthermore, we abbreviate two stacked vectors using $(a, b) := [a^\top, b^\top]^\top$. We also define $\mathcal{B}_\delta(x) := \{\hat{x} \mid \|\hat{x} - x\| \leqslant \delta\}$ as the norm ball of radius δ centered at x. The modulo operator, denoted by $a \bmod b$, returns the remainder of the integer division of a by b, where $a, b \in \mathbb{N}$. A specific time instant is denoted by t, while $k|t$ refers to the internal time of the controller, representing the time k steps ahead of the time instant t. We denote the set of all M agents as $\mathcal{A} := \{1, \dots, M\}$ and label them with $v \in \mathcal{A}$. Each agent v controls the open-loop predictions $x^v_{k|t} \in \mathbb{R}^{n^v}$ and $u^v_{k|t} \in \mathbb{R}^{m^v}$ referring to the state and input vector at time step k given time instant t. We compactly denote the stacked vectors formed by the decision variables of all agents as $\boldsymbol{x}_{k|t} := \left[(x^1_{k|t})^\top, \dots, (x^M_{k|t})^\top\right]^\top \in \mathbb{R}^n$ and $\boldsymbol{u}_{k|t} := \left[(u^1_{k|t})^\top, \dots, (u^M_{k|t})^\top\right]^\top \in \mathbb{R}^m$, with the dimensions $n := \sum_{v=1}^M n^v$ and $m := \sum_{v=1}^M m^v$. Further $\boldsymbol{x}^{-v}_{k|t}$ and $\boldsymbol{u}^{-v}_{k|t}$ refer to the stacked open-loop predictions formed by all agents' decision variables except for agent v and we sometimes write $(x^v_{k|t}, \boldsymbol{x}^{-v}_{k|t})$ instead of $\boldsymbol{x}_{k|t}$ to emphasize the dependency on the v-th agent's variables. Finally we denote the stacked vectors over the horizon by dropping the time index such as $x^v_{\bullet|t}$ and $u^v_{\bullet|t}$ where $\boldsymbol{x}_{\bullet|t}$ and $\boldsymbol{u}_{\bullet|t}$ refer to the stacked vectors $\boldsymbol{x}_{k|t}$ and $\boldsymbol{u}_{k|t}$ stacked over all $k \in \mathbb{N}_{N+1}$ and $k \in \mathbb{N}_N$ respectively. Last but not least we denote the closed-loop trajectory at time t by (x_t, u_t) and $(\boldsymbol{x}_t, \boldsymbol{u}_t)$ representing the trajectory of agent v and the stacked vector of all agents' decision variables, respectively.

I also replaced $\mathbb{N}$ with $\mathbb{N}$ for consistency with standard notation and explicitly stated that $b \neq 0$ to avoid division by zero issues. Let me know if you'd like further refinements!

2.1 Comparison Functions

In the following we define four comparison functions that are commonly used for the convergence analysis of trajectories which classically involves the definition of general lower and upper bounds.

- We say that a function $\gamma : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$ belongs to class $\mathcal{K}$ if it is continuous, strictly *increasing*, and $\gamma(0) = 0$.
- Let $\mathcal{K}_\infty$ represent the set of continuous functions where $\gamma : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$ is considered to be in class $\mathcal{K}_\infty$ if it is a $\mathcal{K}$-function and $\gamma(r) \to \infty$ as $r \to \infty$.

- We say that a function $\gamma : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$ belongs to class $\mathcal{L}$ if it is continuous and strictly *decreasing* with $\lim_{r\to\infty} \gamma(r) = 0$.
- A continuous function $\gamma : \mathbb{R}_{\geqslant 0} \times \mathbb{N}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$ is of class $\mathcal{KL}$ if for each $t \in \mathbb{N}_{\geqslant 0}$, $\gamma(\cdot, t)$ is of class $\mathcal{K}$ and for each $r \in \mathbb{R}_{\geqslant 0}$, $\gamma(r, \cdot)$ is of class $\mathcal{L}$.

In Figure 2.1 below we give an example for some of the above introduced comparison functions.

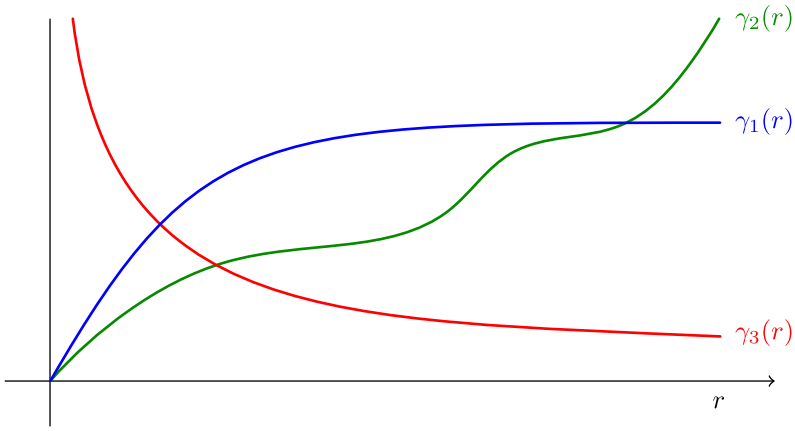

Figure 2.1: Illustration of $\gamma_1(r) \in \mathcal{K}$, $\gamma_2(r) \in \mathcal{K}_\infty$ and $\gamma_3(r) \in \mathcal{L}$.

Chapter 3

Theoretical Background and Supporting Results

In this chapter, we provide some background on game theory and MPC theory, with a specific focus on potential games and EMPC, dicussing results on stability with and without terminal ingredients. Rather than offering a comprehensive summary of these topics, this chapter aims to present a selection of key concepts allowing us to draw theoretical connections between the two topics, as outlined in the Introduction.

3.1 Game Theory

In this section, we provide a basic introduction to game theory, covering the characteristics of a game together with the most important concepts, such as a formal definition of equilibria and their stability. In Subsection 3.1.1 we then delve into potential games, a special class of games that can be reformulated in a way such that they can be linked to EMPC theory.

Game theory is a branch of mathematics that provides a framework to model the interactions of rational, self-interested, decision makers also referred to as players or agents. Its development was largely motivated by the study of micro- and macroeconomics made famous, as early as 1944, by John von Neumann and Oscar Morgenstern with the publication of their groundbreaking work [15] (cf. [1, Foreword]). Nowadays, game theory finds its application not only in economics but across a wide range of different disciplines such as, e.g., biology, social sciences, politics, engineering, etc. Its popularity comes from the fact that many systems that involve humans as decision-makers inherently take the form of a game, where individual choices and interactions collectively shape the the overall system outcome. Refer to [1, 1.1 Games, Old and New] for a collection of a variety of different games, discussing the usefulness of game theory together with its potential areas of application.

Let us formally define a game with M agents, where each agent v controls their *decision variables* $z^v \in \mathcal{Z}^v$, with $\mathcal{Z}^v$ denoting their *action space*, also referred to as *constraints*. A set of decision variables chosen by an agent v is commonly called their *strategy*. Furthermore, we compactly denote the vector formed by the stacked decision variables of all agents as $\boldsymbol{z} := \left[(z^1)^\top, \dots, (z^M)^\top\right]^\top \in \mathbb{R}^n$ where n denotes the total dimension $n := \sum_{v=1}^{M} n^v$. With $\boldsymbol{z}^{-v}$ we denote the stacked vector of all agents' decision variables excluding agent v and we sometimes emphasize the dependence on agent v's decision variables by writing $(z^v, \boldsymbol{z}^{-v})$. It's important to note that this notation is purely representational and does not imply any reordering of the components in $\boldsymbol{z}$. Each agent v aims to

to optimize an *objective function*[1] $\ell^v(z^v, \boldsymbol{z}^{-v})$ that depends on both its own decision variables z^v and the other agents' decision variables $\boldsymbol{z}^{-v}$. Building on the above notation, we now introduce the game, commonly referred to as the Nash equilibrium problem (NEP) [18, Chapter 12.2].

Definition 3.1. A *Nash equilibrium problem* is a coupled optimal control problem (OCP) where each agent v is minimizing their objective $\ell^v(z^v, \boldsymbol{z}^{-v})$ with respect to their own strategy $z^v \in \mathcal{Z}^v$, given the other agents' strategies $\boldsymbol{z}^{-v}$, such that

$$\forall\, v \in \mathcal{A}: \quad \begin{cases} \min\limits_{z^v} & \ell^v(z^v, \boldsymbol{z}^{-v}) \\ \text{s.t.} & z^v \in \mathcal{Z}^v \end{cases} \tag{3.1}$$

where $\mathcal{A} := \{1, \dots, M\}$ denotes the set of all agents v.

A more general form of the NEP is the so called generalized Nash equilibrium problem where the coupling of the OCP is not only introduced through the cost but also through the constraints [16, Equation 1].

Definition 3.2. A *generalized Nash equilibrium problem (GNEP)* is an OCP that is coupled both in the objective and in the constraints, where each agent v seeks to find a strategy $z^v \in \mathcal{Z}^v(\boldsymbol{z}^{-v})$ that minimizes his objective $\ell^v(z^v, \boldsymbol{z}^{-v})$, given the strategies of the other agents $\boldsymbol{z}^{-v}$ expressed by

$$\forall\, v \in \mathcal{A}: \quad \begin{cases} \min\limits_{z^v} & \ell^v(z^v, \boldsymbol{z}^{-v}) \\ \text{s.t.} & z^v \in \mathcal{Z}^v(\boldsymbol{z}^{-v}). \end{cases} \tag{3.2}$$

Hereby $\mathcal{Z}^v(\boldsymbol{z}^{-v})$ represents the constraint set of agent v, which depends on the strategies of the other agents $\boldsymbol{z}^{-v}$ and is defined as

$$\mathcal{Z}^v(\boldsymbol{z}^{-v}) := \left\{ z^v \mid (z^v, \boldsymbol{z}^{-v}) \in \boldsymbol{\mathcal{Z}} \right\} \tag{3.3}$$

with $\boldsymbol{\mathcal{Z}}$ denoting the coupling constraints of all agents' decision variables. This immediately gives rise to the definition of the (generalized) Nash equilibrium (NE) as introduced below and taken from [19, Equation 1].

Definition 3.3. The *generalized Nash equilibrium* $\bar{\boldsymbol{z}}$ is a solution to the GNEP (3.2) where each agent v minimizes its cost $\ell^v(\bar{z}^v, \bar{\boldsymbol{z}}^{-v})$, given the other agents strategies $\bar{\boldsymbol{z}}^{-v}$, such that the following inequality holds for all $y^v \in \mathcal{Z}^v(\bar{\boldsymbol{z}}^{-v})$

$$\ell^v(\bar{z}^v, \bar{\boldsymbol{z}}^{-v}) \leqslant \ell^v(y^v, \bar{\boldsymbol{z}}^{-v}). \tag{3.4}$$

By definition, the generalized Nash equilibrium (GNE) is strategically stable, i.e. no agent involved can gain any advantage by unilaterally changing their strategy.

Remark 1. GNEs are not necessarily unique, in fact, there can potentially be infinitely many. Moreover, the objective value achieved at different GNEs can vary, implying that some equilibria might be more desirable (for some agents) than others in terms of the achieved objective.

A more detailed discussion of GNEPs is available in [16], while [1] provides an overview of game theory applications in engineering along with an introduction to the field.

3.1.1 Potential Games

Potential games are a specific class of games in which agents unknowingly optimize a shared objective, known as the potential function, which exists only under a symmetric coupling between

[1] The objective function is sometimes also referred to as cost function, utility function, payoff function, loss function or profit function [16, 17].

the agents [17]. This potential function allows to reformulate a GNEP as defined in (3.2) into a centralized OCP with the potential function as the objective.

We start by defining the class of generalized ordinal potential games according to the definition in [20, Definition 2.4].

Definition 3.4. A GNEP is a *generalized ordinal potential game* if there exists a continuous function $p(\boldsymbol{z})$ such that the following relation holds for all $v \in \mathcal{A}$

$$\ell^v(z^v, \boldsymbol{z}^{-v}) - \ell^v(y^v, \boldsymbol{z}^{-v}) < 0 \;\Rightarrow\; p(z^v, \boldsymbol{z}^{-v}) - p(y^v, \boldsymbol{z}^{-v}) < 0 \qquad \forall\, y^v, z^v \in \mathcal{Z}^v(\boldsymbol{z}^{-v}) \tag{3.5}$$

with $\boldsymbol{z}^{-v}$ such that $\mathcal{Z}^v(\boldsymbol{z}^{-v}) \neq \emptyset$.

A special case of the above definition is the generalized weighted potential game (w-PG) [20, Definition 2.2] defined as follows.

Definition 3.5. A GNEP is a *weighted generalized potential game* if there exists a continuous function $p(\boldsymbol{z})$ such that the following relation holds for all $v \in \mathcal{A}$

$$\ell^v(z^v, \boldsymbol{z}^{-v}) - \ell^v(y^v, \boldsymbol{z}^{-v}) = w^v\big(p(z^v, \boldsymbol{z}^{-v}) - p(y^v, \boldsymbol{z}^{-v})\big) \qquad \forall\, y^v, z^v \in \mathcal{Z}^v(\boldsymbol{z}^{-v}). \tag{3.6}$$

Remark 2. A w-PG is called an *exact potential* game if $w^v = 1$ for all $v \in \mathcal{A}$.

The existence of a potential function as defined in (3.5) and (3.6) enables us to reformulate the GNEP (3.2) into a centralized problem. Specifically, by replacing the agents' objective with the potential function $p(\boldsymbol{z})$ [19, Equation 7] we get to the OCP

$$\begin{aligned} \min_{\boldsymbol{z}} \quad & p(\boldsymbol{z}) \\ \text{s.t.} \quad & \boldsymbol{z} \in \boldsymbol{\mathcal{Z}} \end{aligned} \tag{3.7}$$

where the coupling constraints are denoted by $\boldsymbol{\mathcal{Z}}$ as defined in (3.3). This result plays an important roll in connecting PGs to EMPC theory.

A natural question that arises from the reformulation in (3.7) is whether this problem admits an equilibrium and, if so, whether its equilibria coincide with those of the decentralized formulation (3.2). According to [20, Theorem 2.3], an (ordinal) PG admits at least one pure-strategy NE if $\boldsymbol{\mathcal{Z}}$ is compact (i.e., closed and bounded) and the potential function $p(\boldsymbol{z})$ is continuous. Moreover, if the objective is strictly convex for all agents $v \in \mathcal{A}$, this NE is also unique. Furthermore, [17, Lemma 2.1] shows that the NEs of a NEP coincide with those of its centralized formulation (3.7) when no coupling constraints are present. However, in the case of GNEPs, where coupling constraints are present, the equilibria of (3.7) generally form a subset of those in (3.2), even if (3.7) is convex (cf. [19, Section 3.2],[21]).

In other words, we can always expect (3.7) to have at least one NE, which is unique if the objective is strictly convex for all agents $v \in \mathcal{A}$.

The next result is taken from [20, Definition 2.2, Equation 2.4] and provides a method to construct w-potential functions for continuously differentiable objectives.

Lemma 3.1. *Let the objective function $\ell^v(z^v, \boldsymbol{z}^{-v})$ be continuously differentiable for all $v \in \mathcal{A}$. Then, a w-potential function $p(\boldsymbol{z})$ satisfies the following condition*

$$\frac{\partial\, \ell^v(z^v, \boldsymbol{z}^{-v})}{\partial\, z^v} = w^v \frac{\partial\, p(z^v, \boldsymbol{z}^{-v})}{\partial\, z^v} \tag{3.8}$$

for all $v \in \mathcal{A}$.

An application of this lemma can be found in the proof of Theorem 4.2.

The table on the following page summarizes potential functions commonly used in engineering applications. For a general introduction to potential games, see [20].

Table 3.1: Different potential functions used in engineering applications

Objective $\ell^v(z^v, \boldsymbol{z}^{-v})$	**Potential Function** $p(\boldsymbol{z})$	**Application Areas**	**Papers**
$c(\boldsymbol{z}) + d^v(z^v)$	$c(\boldsymbol{z}) + \sum_{v=1}^{M} d^v(z^v)$	Autonomous Driving, Multi-Agent Trajectory Optimization, Traffic Routing	[8, 6, 5]
$d^v(z^v) + \sum_{i \in \mathcal{A} \setminus v} k^{v,i}(z^v, z^i), \quad (k^{v,i} = k^{i,v})$	$\sum_{v=1}^{M} d^v(z^v) + \frac{1}{2} \sum_{i \in \mathcal{A} \setminus v} k^{v,i}(z^v, z^i)$	Smart e-Mobility, Transportation Economics, Urban Driving Games	[10, 11]
$d^v(z^v) + \sum_{i=1}^{M} (z^v)^\top C^{v,i} z^i, \ (C^{v,i} = (C^{i,v})^\top)$	$\sum_{v=1}^{M} d^v(z^v) + \frac{1}{2} \left((z^v)^\top C^{v,v} z^v + \sum_{i=1}^{M} (z^v)^\top C^{v,i} z^i \right)$	Demand-Side Management, Battery-Charging Game, Network Congestion Control	[7, 4, 9, 5]

Legend

- $d^v(z^v) \in \mathbb{R}$ continuous, nonlinear objective function only depending on agent v
- $c(z^v, \boldsymbol{z}^{-v}) \in \mathbb{R}$ continuous, nonlinear objective function depending on all agents $v \in \mathcal{A}$
- $k^{i,j}(z^i, z^j) \in \mathbb{R}$ with $\left(k^{i,j} = k^{j,i}\right)$ continuous, nonlinear objective function depending on agent i and j (symmetric)
- $C^{i,j} \in \mathbb{R}^{n^i \times n^j}$ with $\left(C^{i,j} = (C^{j,i})^\top\right)$ continuous, quadratic objective function depending on agent i and j (symmetric)

3.2 Economic Model Predictive Control

In this section we give a brief introduction on MPC and EMPC followed by an in-depth discussion on how to achieve stability and recursive feasibility with and without terminal ingredients for EMPC. Specifically we will discuss two methods proposed by [13] and [14] which later find their application in Chapter 4.

Model predictive control is an optimization-based control scheme that continuously reacts to its environment using a system model, also referred to as dynamics, to predict the optimal response over a limited prediction horizon N. Thereby, it iteratively updates its decision at each time step, enabling it to adapt to model mismatches and noise (see Figure 3.1). Originally it was developed for process control in in sectors such as chemical processing and oil refining and has since gained increasing adoption in a variety of other area application areas, driven by the improving computational power of micro controllers [22]. The main advantage of optimization-based control is the ability to explicitly consider constraints while optimizing a given objective [3], a feature that most classical control methods lack. Over the years, a vast body of literature has emerged, proposing various MPC formulations for both discrete and continuous time systems that extend far beyond nominal steady-state tracking. These include robust [23, 24, 25, 26, 27] and stochastic MPC [28, 29, 30] to handle noise and model mismatches, learning-based MPC for online adaptation [24, 31, 32, 33, 34, 35], safety filters for ensuring safe operation of learning-based methods [36, 37, 38, 39, 40] and EMPC to handle general economic costs that are nonconvex and nonzero at the steady-state [41, 42, 43, 44, 14, 13, 45, 46]. Certainly, this list is non-exhaustive, as various other methods exist, some of which combine the approaches mentioned here. A broader perspective on MPC frameworks is given by the work of [47], while a rather comprehensive treatment of the topic can be found in [3].

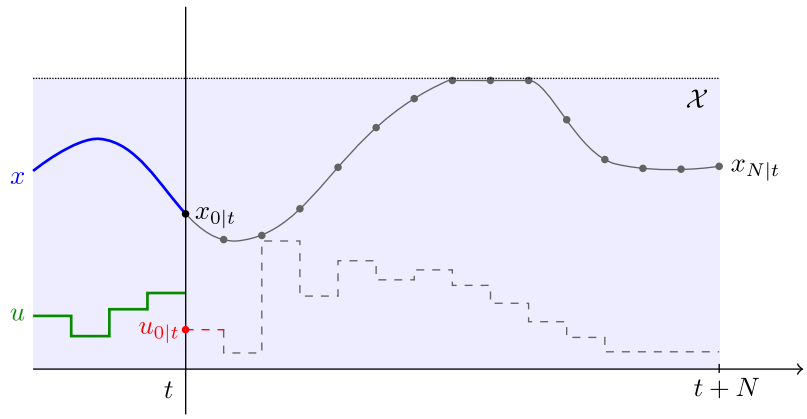

Figure 3.1: The closed-loop state trajectory (blue) and input trajectory (green) are shown alongside the open-loop state predictions $x_{k|t}$ (solid gray) and the discrete input predictions $u_{k|t}$ (dashed gray) over a horizon N at time t. The shaded area indicates the state constraints, and the red line highlights the next input $u_{0|t}$ applied to the system, emphasizing the iterative nature of MPC.

In the following, we focus on nominal discrete-time (economic) MPC without terminal ingredients. To that end, we consider the discrete-time, time-varying, nonlinear dynamics governed by

$$x_{t+1} = f_t(x_t, u_t), \tag{3.9}$$

subject to the time-invariant hard state and input constraints

$$(x_t, u_t) \in \mathcal{X} \times \mathcal{U} \tag{3.10}$$

with $\mathcal{X}$ and $\mathcal{U}$ denoting some nonempty sets.

We further define the time-varying objective function

$$J_N\big(x_{\cdot|t}, u_{\cdot|t}, t\big) := \sum_{k=0}^{N-1} \ell_{t+k}(x_{k|t}, u_{k|t}) \tag{3.11}$$

which represents the sum of the time-varying stage costs $\ell_t(x_t, u_t)$.

Remark 3. The stage cost $\ell_t(x_t, u_t)$ is what distinguishes MPC from EMPC. In standard MPC, we typically assume a continuous, positive definite stage cost that is zero at a steady-state[2] (x_s, u_s), with $\ell_t(x_t, u_t)$ being lower and upper bounded by quadratic functions.

As previously established in Chapter 2, we denote the closed-loop state trajectory at time t by x_t, with u_t representing the closed-loop input trajectory. The open-loop predictions of the MPC controller at time t are denoted by $x_{k|t}$ and $u_{k|t}$ for the state and input, respectively.

Given the dynamics, objective and constraints defined above, the (economic) MPC control law is obtained by recursively applying the first input $u_{0|t}$ obtained from solving the following OCP

$$\begin{aligned} V_N(x(t), t) := \min_{x_{\cdot|t}, u_{\cdot|t}} \quad & J_N\big(x_{\cdot|t}, u_{\cdot|t}, t\big) \\ \text{s.t.} \quad & x_{0|t} = x(t), \\ & x_{k+1|t} = f_{t+k}(x_{k|t}, u_{k|t}), && \forall\, k \in \mathbb{N}_N \\ & x_{k|t} \in \mathcal{X}, && \forall\, k \in \mathbb{N}_{N+1} \\ & u_{k|t} \in \mathcal{U} && \forall\, k \in \mathbb{N}_N \end{aligned} \tag{3.12}$$

where $x(t)$ denotes the initial state at time instance t, with $V_N(x(t), t)$ being the optimal value function.

Since the MPC formulation in (3.12) lacks terminal ingredients, recursive feasibility is not inherently guaranteed. Recursive feasibility ensures that once the MPC problem is initialized at a feasible point, it remains feasible at all future time steps. Establishing recursive feasibility without terminal ingredients requires specific system properties such as Turnpike and a suitable controllability assumption. Another key challenge is the asymptotic stability of the proposed scheme. Due to the potential nonconvexity of the cost function, the closed-loop system may not naturally regulate to a steady-state, making stability guarantees nontrivial. Section 3.2.2 examines specific system properties that ensure both asymptotic stability and recursive feasibility. In Section 3.2.3 we alternatively discuss how to derive recursive feasibility and asymptotic stability with an exact terminal constraint and the additional assumption of dissipativity.

3.2.1 Definitions

Before stating specific results from EMPC theory, we first introduce some commonly used definitions that are used to prove stability and recursive feasibility.

We start with a notation of optimality as first proposed by [48]. This concept is introduced to address the challenges posed by the infinite-horizon OCP, in particular the problem of the potentially unbounded value function J_∞. This unboundedness violates the standard definition of optimality, as it results in the absence of a unique minimum value and can be resolved by considering the difference in stage cost ℓ_t as defined below.

Definition 3.6. A state and input trajectory $(x^\star, u^\star)$ is called *overtaking optimal* if

$$\liminf_{N\to\infty} \sum_{t=0}^{N-1} \ell_t(x_t, u_t) - \ell_t(x_t^\star, u_t^\star) \geqslant 0. \tag{3.13}$$

[2] A steady-state refers to a state-input pair (x_s, u_s) that satisfies the condition $x_s = f(x_s, u_s)$, i.e., the system stays at the steady-state as long as u_s is applied.

General overtaking optimality is defined relative to a trajectory. When the system is optimal with respect to a steady-state pair, the trajectory simplifies to the time-invariant pair (x_s, u_s).

Inspired by overtaking optimality we further introduce the shifted stage cost and the associated shifted cost and value function, as defined in [49, Definition 3].

Definition 3.7. The *shifted stage* cost is defined relative to a reference trajectory (x^r, u^r) given by

$$\hat{\ell}_t(x_t, u_t) := \ell_t(x_t, u_t) - \ell_t(x_t^r, u_t^r) \tag{3.14}$$

and is denoted as $\hat{\ell}_t^\star(x_t, u_t)$ if defined relative to the overtaking optimal pair $(x^\star, u^\star)$. The *shifted cost*, which follows from the shifted stage cost, is defined as

$$\hat{J}_N(x, u, t) = \sum_{k=t}^{t+N-1} \hat{\ell}_k(x_k, u_k) \tag{3.15}$$

with its corresponding *shifted value function* $\hat{V}_N(x(t), t)$.

When clear from context, we drop the superscript $\bullet^\star$ and just refer to the shifted stage cost, cost and value function without it.

Following [49, Definition 7], we introduce two notions of dissipativity, providing a framework to assess the stability of our OCP based on the stage cost. Dissipativity can also be understood as a notion of energy that is stored in a system [50].

Definition 3.8. A system is *strictly dissipative* with respect to a reference trajectory (x_t^r, u_t^r) and the supply rate $s_t(x_t, u_t)$ if there exists a time-varying storage function $\lambda_t(x_t) \in \mathbb{R}$ bounded from below, together with $\alpha(\|(x_t - x_t^r, u_t - u_t^r)\|) \in \mathcal{K}_\infty$ such that

$$\lambda_{t+1}(f(x_t, u_t)) - \lambda_t(x_t) \leqslant -\alpha(\|(x_t - x_t^r, u_t - u_t^r)\|) + s_t(x_t, u_t) \tag{3.16}$$

holds for all $t \in \mathbb{N}$ and $(x_t, u_t) \in \mathcal{X} \times \mathcal{U}$. The supply rate is defined as

$$s_t(x_t, u_t) := \ell_t(x_t, u_t) - \ell_t(x_t^r, u_t^r) \overset{(3.14)}{=} \hat{\ell}_t(x_t, u_t). \tag{3.17}$$

A less restrictive version of strict dissipativity is *dissipativity* which is given for $\alpha = 0$.

Remark 4. The above definition of dissipativity is sometimes also referred to as pre-dissipativity (cf. [51, 52, 45]) since it does not hold globally but only on the set $\mathcal{X} \times \mathcal{U}$. For simplicity, we will refer to dissipativity valid on the entire set $\mathbb{R}^n \times \mathbb{R}^m$ as *global dissipativity*.

Following [13, Definition 11] we define the rotated stage cost, cost and value function as a consequence of the above definition of dissipativity.

Definition 3.9. The *rotated stage cost* is given by

$$\tilde{\ell}_t(x_t, u_t) = \hat{\ell}_t(x_t, u_t) + \lambda_t(x_t) - \lambda_{t+1}(f_t(x_t, u_t)) \tag{3.18}$$

with the corresponding *rotated cost* defined as

$$\tilde{J}_N(x, u, t) = \sum_{k=t}^{t+N-1} \tilde{\ell}_t(x_k, u_k) = \hat{J}_N(x, u, t) + \lambda_0(x_t) - \lambda_{t+N}(x_{t+N}) \tag{3.19}$$

and the *rotated value function* denoted by $\tilde{V}_N(x(t), t)$.

Below we present the Turnpike property according to the definition in [53, Proposition 4.1], which is a system property that is normally used to show stability for EMPC without terminal constraints.

Definition 3.10. Consider a trajectory pair $(x^\star, u^\star)$ at which the system (3.12) is optimally operated. We say that an OCP has the *Turnpike property* if there exists $C < \infty$ and $\alpha(\epsilon) \in \mathcal{K}_\infty$ such that

$$\#\mathcal{Q}_\epsilon \geqslant N - \frac{C}{\alpha(\epsilon)} \tag{3.20}$$

with $\mathcal{Q}_\epsilon := \{t \in \mathbb{N}_N \mid (x_t, u_t) \in \mathcal{B}_\epsilon(x_t^\star, u_t^\star)\}$ the set of all times t where the trajectory lies within an ϵ-vicinity of the overtaking optimal pair $(x^\star, u^\star)$. $\#\mathcal{Q}_\epsilon$ denotes the cardinality of $\mathcal{Q}_\epsilon$, which corresponds to the total time spent *within this ϵ-vicinity* around the optimal trajectory.

Remark 5. This definition is equivalent to the one presented in [13, Definition 6] with the difference that they define the set

$$\mathcal{R}_P = \mathbb{N}_N \setminus \mathcal{Q}_\epsilon \tag{3.21}$$

which describes the total time spent *outside of the ϵ-vicinity* around the optimal trajectory, such that one can write

$$\#\mathcal{R}_P = N - \#\mathcal{Q}_\epsilon \leqslant \frac{C}{\alpha(\epsilon)} = P \tag{3.22}$$

with the constant $P \in \mathbb{N}$ and hence $\epsilon = \alpha^{-1}\left(\frac{C}{P}\right) = \sigma(P) \in \mathcal{L}$.

The Turnpike property implies that the open-loop trajectory of the OCP will stay close to the optimal trajectory $(x^\star, u^\star)$ for a nonzero amount of time until it diverges again as depicted in Figure 3.2.

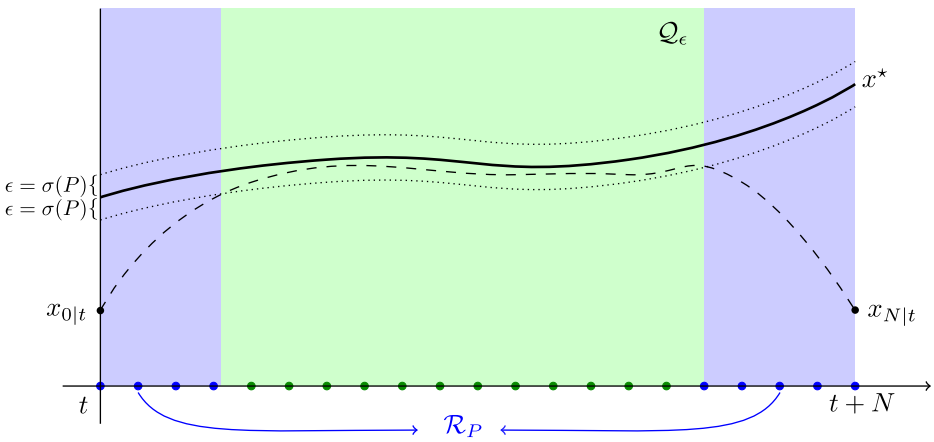

Figure 3.2: The Turnpike property according to the definition from [53] in green and [13] blue (Figure adapted from [54, Figure 4.2]).

3.2.2 Economic MPC without Terminal Ingredients

In this section, we wish to establish conditions under which we can show asymptotic stability and recursive feasibility for EMPC without terminal ingredients. This is desirable since it gives us insights about stability for the most general form of our game, without enforcing any additional (artificial) measures to provide stability.

An important step toward developing a stability theory for EMPC without terminal ingredients was made in [55], which built on the recent insights from [56, 57] regarding the role of dissipativity in economic model predictive control. Therein, they prove near optimal performance and convergence to the optimal steady-state for a discrete time, nonlinear system with a time-invariant stage cost, using the Turnpike property and a suitable controllability assumptions. In [43], the concept of optimal steady-state behavior is generalized to optimal periodic behavior for discrete-time, time-invariant nonlinear systems with a time-invariant stage cost. The authors propose a multi-step MPC scheme, from which they derive performance guarantees and establish convergence to the optimal periodic orbit based on dissipativity and controllability assumptions. The special case of discrete time, linear time-invariant (LTI) systems and linear quadratic stage cost is discussed in [51] where they derive necessary and sufficient conditions for the Turnpike property to hold. They further introduce a new property referred to as pre-dissipativity (see 4). The authors of [58] suggest a

robust EMPC formulation without terminal ingredients for a specific class of time-varying systems, given a time-invariant stage cost, introducing a new definition of robust dissipativity. They prove the Turnpike property for this new dissipativity formulation but give no stability or performance guarantees. The work of [59] establishes practical asymptotic stability with respect to an unknown optimal periodic orbit together with performance guarantees for a discrete time, time-invariant nonlinear system and and a discounted, time-invariant stage cost. Recently [46], inspired by robust MPC, analyzed EMPC using the notion of local incremental stabilizability being able to rederive most of the important results for discrete time, time-invariant nonlinear systems and stage costs such as practical asymptotic stability and an averaged closed-loop performance bound.

To establish stability without terminal ingredients, we rely on theorems presented in [13]. The theorems therein guarantee that EMPC produces initial segments of near-optimal infinite-horizon trajectories, further showing that the optimal infinite-horizon trajectory exhibits practical asymptotic stability as defined in [60]. While these results are applicable to general nonlinear, time-varying dynamics, we focus here on the specific case of linear time-varying (LTV) dynamics, for which, alternative assumptions are provided in [49], partially replacing those from [13]. The LTV dynamics are defined as follows

$$f_t(x,u) = A_t x + B_t u + d_t. \tag{3.23}$$

We further assume that A_t is full rank (i.e., invertible) for all $t \in \mathbb{N}$. This condition is typically met in discretized continuous-time systems because discretization methods, such as the Euler method, generally produce matrices with full rank.

The state space $\mathcal{X}$ is a nonempty, bounded, open subset of $\mathbb{R}^n$ and the system is subject to the convex input constraints

$$u_t \in \mathcal{U} := \big\{u_t \in \mathbb{R}^m \mid h_u(u_t) \leqslant 0\big\} \tag{3.24}$$

where $h_u(u_t) \in \mathbb{R}^{c_u}$ is a continuous, differentiable function.

Remark 6. State constraints cannot currently be considered due to the reliance on the strong maximum principle (Theorem 3.3), which is essential for establishing dissipativity in general time-varying systems. At present, no discrete-time infinite-horizon Pontryagin principle accommodates both state and input constraints simultaneously without requiring very strong assumptions such as inactive constraints along the optimal trajectory. Infinite-horizon theorems are normally derived using a reduction to a sequence of finite-horizon problems [61, 1.3 The Reduction to the Finite Horizon]. These reduced problems can than be analyzed using finite-horizon theorems, also referred to as multiplier rules (cf. [61, Chapter 1.4.2-1.4.4]), by extending the horizon back to infinity. However, this extension is only meaningful if the associated multipliers remain bounded for all times (cf. [62, 5. Proof of Theorem 3.1, (5.15)], proof of [61, Theorem 2.19]). Proving this boundedness of the Lagrange multipliers provided by the multiplier rules remains a significant challenge once state and input constraints are present, often requiring very strong assumptions (cf. [61, Theorem 2.19, Condition (3)], [61, Theorem 2.20, Condition (4)]).

The conditions for asymptotic stability will be established in the next section, followed by a result on recursive feasibility.

Stability

In order to prove asymptotic stability without terminal constraints we assume uniform strict convexity according to the definition in [49, Assumption 5].

Assumption 3.2. [Uniform Strict Convexity] There exists $\kappa > 0$ and $\gamma\big(\|(x_1 - x_2, u_1 - u_2)\|\big) \in \mathcal{K}_\infty$ such that

$$\begin{aligned}\ell_t\big(\theta x_1 + (1-\theta)x_2,\, \theta u_1 + (1-\theta)u_2\big) \leqslant\; & \theta \ell_t\big(x_1, u_1\big) + (1-\theta)\ell_t\big(x_2, u_2\big) \\ & - \kappa\,\theta(1-\theta)\gamma\big(\|(x_1 - x_2, u_1 - u_2)\|\big)\end{aligned} \tag{3.25}$$

holds for all $t \in \mathbb{N}$, $(x_1, u_1), (x_2, u_2) \in \mathcal{X} \times \mathcal{U}$ and $\theta \in [0, 1]$.

To present the next theorem, we introduce the following condition, which we will use instead of Michel's convexity condition [63, Condition (4.1)] as proposed in [63, Condition (4.2)]. While the two conditions are equivalent, Michel's convexity condition is set-based and less intuitive than the condition presented below.

Definition 3.11. The *Ioffe-Tihomirov condition* states that for all $t \in \mathbb{N}$, for all $x \in \mathcal{X}$, for all $u_1, u_2 \in \mathcal{U}$, and for all $\theta \in [0,1]$, there exists $u \in \mathcal{U}$ such that the following three relations hold

$$\begin{aligned} \ell_t(x,u) &\leqslant \theta \ell_t(x,u_1) + (1-\theta)\,\ell_t(x,u_2) \\ f_t(x,u) &= \theta f_t(x,u_1) + (1-\theta)\,f_t(x,u_2) \\ h_u(x,u) &\leqslant \theta h_u(x,u_1) + (1-\theta) h_u(x,u_2)\,. \end{aligned}$$

Remark 7. Note that the above conditions are readily satisfied given our LTV dynamics, the convex input constraints and the uniform strict convexity provided by Assumption 3.2.

We also define the pre-Lagrangian, a fundamental component in deriving optimality conditions.

Definition 3.12. The *pre-Lagrangian* $L_t(x_t, u_t, q_{t+1}, \eta_t, \mu_t) \in \mathbb{R}$ of problem (3.12) is defined as

$$L_t(x_t, u_t, q_{t+1}, \eta_t, \mu_t) := \eta_t\, \ell_t(x_t, u_t) + q_{t+1}^\top f_t(x_t, u_t) + \mu_t^\top h_u(u_t) \tag{3.26}$$

where $\eta_t \in \mathbb{R}$, $q_{t+1} \in \mathbb{R}^n$ and $\mu_t \in \mathbb{R}^c$ are the Lagrange multipliers.

Below we restate the strong strong maximum principle from [63, Theorem 4.1], adapted to our notation, which poses a necessary condition for optimality of the infinite-horizon OCP. It is similar to the one used in [49, Theorem 3] but allows for input constraints.

Theorem 3.3. *Let $(x^\star, u^\star)$ be an overtaking optimal pair for the infinite-horizon counter part of the general OCP* (3.12) *subject to the constraints defined in* (3.24)*. If it holds that*

(1) *For all $t \in \mathbb{N}$, $\mathcal{X}$ is nonempty open subset of $\mathbb{R}^n$, and $h_u(x_t, u_t) \in \mathbb{R}^{c_u}$.*

(2) *For all $t \in \mathbb{N}_N$, the partial differential $\frac{\partial f_t}{\partial x}\Big|_{(x_t^\star, u_t^\star)}$ is invertible.*

(3) *For all $t \in \mathbb{N}_N$, the Ioffe-Tihomirov condition holds, given in 3.11.*

Then, there exist $\eta_0 \in \mathbb{R}$, $q_{t+1} \in \mathbb{R}^n$ and $\mu_t \in \mathbb{R}^{c_u}$ for all $t \in \mathbb{N}$ satisfying the following conditions

(i) $(\eta_0, q_1) \neq (0,0)$.

(ii) $\eta_0 \geqslant 0$, $\mu_t \geqslant 0$ *and* $\mu_t^\top h_u(u_t^\star) = 0$ *for all* $t \in \mathbb{N}$.

(iii) *For all $t \in \mathbb{N}$, it holds*

$$q_t^\top = \frac{\partial L_t}{\partial x}\Big|_{(x_t^\star, u_t^\star, q_{t+1}, \eta_0)}.$$

(iv) *For all $t \in \mathbb{N}$, it holds*

$$u_t^\star = \arg\min_{u \in \mathcal{U}} \; L_t(x_t^\star, u, q_{t+1}, \eta_0, \mu_t).$$

To simplify further analysis, we establish the following lemma, which verifies that the conditions required by Theorem 3.3 hold and state an alternative first-order necessary condition for Condition (iv) which we later use in the proof of Theorem 3.5 in combination with the Karush-Kuhn-Tucker (KKT) conditions.

Lemma 3.4. *Given the LTV dynamics 3.23, the constraints defined in 3.24 and under Assumption 3.2 there exist $\eta_0 \in \mathbb{R}$, $q_{t+1} \in \mathbb{R}^n$ and $\mu_t \in \mathbb{R}^{c_u}$ for all $t \in \mathbb{N}$ satisfying the following conditions*

(i) $(\eta_0, q_1) \neq (0,0)$.

(ii) $\eta_0 \geqslant 0$, $\mu_t \geqslant 0$ *and* $\mu_t^\top h_u(u_t^\star) = 0$ *for all* $t \in \mathbb{N}$.

(iii) *For all $t \in \mathbb{N}$, it holds*

$$q_t^\top = q_{t+1}^\top \frac{\partial f_t}{\partial x}\bigg|_{(x_t^\star, u_t^\star)} + \eta_0 \frac{\partial \ell_t}{\partial x}\bigg|_{(x_t^\star, u_t^\star)}.$$

(v) *For all $t \in \mathbb{N}$, it holds*

$$0 = q_{t+1}^\top \frac{\partial f_t}{\partial u}\bigg|_{(x_t^\star, u_t^\star)} + \eta_0 \frac{\partial \ell_t}{\partial u}\bigg|_{(x_t^\star, u_t^\star)} + \mu_t^\top \frac{\partial h_u}{\partial u}\bigg|_{(u_t^\star)}.$$

Proof. First note that Condition (v) of Lemma 3.4 is a simple first-order necessary condition of optimality for Condition (iv) of Theorem 3.3 [63, Remark 4.3]. Moreover, the state and input constraints defined in (3.24) satisfy Condition (1) of Theorem 3.3. The LTV dynamics defined in (3.9) further fulfill Condition (2) since we assume that A_t is invertible for all $t \in \mathbb{N}$. The Ioffe-Tihomirov Condition (3) also hold since we have affine dynamics, convex input constraints and uniform strict convexity according to Assumption 3.2 (see Remark 7). □

The next theorem is introduced in [49, Theorem 4] with the difference that we prove it using Lemma 3.4 which allows for input constraints.

Theorem 3.5. *Let Assumption 3.2 hold. If $\eta_0 \neq 0$ then the optimal control problem 3.12 is strictly dissipative on every bounded set $\mathcal{X}$ for all $t \in \mathbb{N}$ with respect to the supply rate $s_t(x_t, u_t) = \hat{\ell}_t^\star(x_t, u_t)$ and the overtaking optimal pair $(x_t^\star, u_t^\star)$.*

Proof. This proof is an adaption of the proof in [49, Theorem 4] for OCPs with input constraints. In order to show strict dissipativity with respect to the supply rate $s_t(x_t, u_t) = \hat{\ell}_t^\star(x_t, u_t)$ one needs to find a storage function $\lambda_t(x_t)$ such that the inequality from Definition 3.8 holds for all $t \in \mathbb{N}$. To that end we propose the following candidate storage function

$$\lambda_t(x_t) = -\frac{q_t^\top}{\eta_0}(x_t - x_t^\star) \tag{3.27}$$

exploiting the structure provided by the solution of the infinite horizon optimal-overtaking trajectory as proposed in [49]. Hereby $q_t^\top$ and η_0 are the Lagrange multipliers of the infinite-horizon OCP provided by Lemma 3.4. Plugging this Ansatz and our storage function into equation (3.16) we write

$$\begin{aligned}\bar{\ell}_t(x_t, u_t) = \eta_0\, \tilde{\ell}_t^\star(x_t, u_t) = \eta_0\, \hat{\ell}_t^\star(x_t, u_t) + q_{t+1}^\top (f(x_t, u_t) - x_{t+1}^\star) - q_t^\top (x_t - x_t^\star) \\ \overset{?}{\geqslant} \eta_0\, \alpha(\|(x_t - x_t^\star, u_t - u_t^\star)\|) = \bar{\alpha}(\|(x_t - x_t^\star, u_t - u_t^\star)\|)\end{aligned} \tag{3.28}$$

where $\bar{\ell}_t(x_t, u_t)$ represents the scaled, rotated stage cost, and $\bar{\alpha}(\|(x_t - x_t^\star, u_t - u_t^\star)\|) \in \mathcal{K}_\infty$ denotes the scaled $\mathcal{K}_\infty$-function. Since we assume $\eta_0 \neq 0$, and by applying Condition (ii) of Lemma 3.4, we conclude that $\eta_0 > 0$, making this scaling unproblematic.

In a next step, we show that $\bar{\ell}_t(x_t, u_t)$ can be lower-bounded by $\bar{\alpha}(\|(x_t - x_t^\star, u_t - u_t^\star)\|)$. First, we demonstrate that $\bar{\ell}_t(x_t, u_t)$ attains its unique minimum at the overtaking-optimal pair $(x^\star, u^\star)$, where $\bar{\ell}_t(x_t^\star, u_t^\star) = 0$ for all times, then we derive the desired lower bound $\bar{\alpha}(\|(x_t - x_t^\star, u_t - u_t^\star)\|)$.

The minimum of $\bar{\ell}_t(x_t, u_t)$ is obtained as a result of the following constraint optimization problem which can alternatively be solved through its dual problem

$$\text{Primal Problem} \begin{cases} \min\limits_{x_t, u_t} & \bar{\ell}_t(x_t, u_t) \\ \text{s.t.} & (x_t, u_t) \in \mathcal{X} \times \mathcal{U}. \end{cases} \Leftrightarrow \text{Dual Problem} \begin{cases} \min\limits_{x_t, u_t, \mu_t} & L(x_t, u_t, \mu_t) \\ \text{s.t.} & \mu_k \geqslant 0 \end{cases} \tag{3.29}$$

with the Lagrangian given by

$$L(x_t, u_t, \mu_t) := \bar{\ell}_t(x_t, u_t) + \mu_t^\top h_u(u_t) \tag{3.30}$$

and its Lagrange multiplier $\mu_t \in \mathbb{R}^{c_u}$. Let's first note that $\ell_t(x_t, u_t)$ is strictly convex by Assumption 3.2, which makes the scaled, rotated stage cost $\bar{\ell}_t(x_t, u_t)$ strictly convex as well, since it is the sum of convex and affine functions (see, e.g., [64, Lemma 2.1.6]). Further, we consider convex constraints $\mathcal{X} \times \mathcal{U}$, which ensures that OCP (3.29) is a convex problem. Consequently, sufficient conditions for $(x_t^\star, u_t^\star)$ to be primal and dual optimal for (3.29) are provided by the KKT conditions (cf. [65, Chapter 5.5.3]) as presented below

$$\begin{aligned} \text{Primal Feasibility}\{ \quad & h_u(u_t) \leqslant 0, \\ \text{Complementary Slackness}\{ \quad & \mu_t^\top h_u(u_t) = 0, \\ \text{Dual Feasibility}\{ \quad & \mu_t \geqslant 0, \\ \text{Stationarity} \begin{cases} \\ \\ \end{cases} & \begin{cases} \left.\dfrac{\partial L}{\partial x_t}\right|_{(x_t,u_t)} = 0, \\ \left.\dfrac{\partial L}{\partial u_t}\right|_{(x_t,u_t)} = 0. \end{cases} \end{aligned} \tag{3.31}$$

Let's begin by demonstrating that the stationarity condition of the Lagrangian associated with (3.29) vanishes at the optimal pair $(x_t^\star, u_t^\star)$

$$\left.\frac{\partial L_t}{\partial x}\right|_{(x_t,u_t)} = \eta_0 \left.\frac{\partial \hat{\ell}_t}{\partial x}\right|_{(x_t,u_t)} + q_{t+1}^\top \left.\frac{\partial f_t}{\partial x}\right|_{(x_t,u_t)} - q_t \tag{3.32}$$

$$\left.\frac{\partial L_t}{\partial u}\right|_{(x_t,u_t)} = \eta_0 \left.\frac{\partial \hat{\ell}_t}{\partial u}\right|_{(x_t,u_t)} + q_{t+1}^\top \left.\frac{\partial f_t}{\partial u}\right|_{(x_t,u_t)} + \mu_t^\top \left.\frac{\partial h_u}{\partial u}\right|_{(x_t,u_t)}. \tag{3.33}$$

By plugging in the overtaking optimal pair $(x_t^\star, u_t^\star)$ and the condition (iii) and (v) from Lemma 3.4 we get

$$\left.\frac{\partial L_t}{\partial x}\right|_{(x_t^\star,u_t^\star)} = 0 \quad \text{and} \quad \left.\frac{\partial L_t}{\partial u}\right|_{(x_t^\star,u_t^\star)} = 0. \tag{3.34}$$

The remaining three KKT conditions are fulfilled with with condition (i) and (ii), hence we have shown that the unique minimizer of $\bar{\ell}(x_t, u_t)$ is given by the overtaking optimal pair $(x_t^\star, u_t^\star)$ for all $t \in \mathbb{N}$. For the reader's convenience, we restate the remainder of the proof, which follows exactly as in [49, Theorem 4].

Consider an arbitrary point $(x_t, u_t) \in \mathcal{X} \times \mathcal{U}$ at some arbitrary time instance $t \in \mathbb{N}$ given some arbitrary $\theta \in (0, 1)$. Let $(\bar{x}_t, \bar{u}_t) := \theta(x_t, u_t) + (1-\theta)(x_t^\star, u_t^\star) \in \mathcal{X} \times \mathcal{U}$ define the convex combination of the pair (x_t, u_t) and the optimal pair $(x_t^\star, u_t^\star)$. By Assumption 3.2 we have that

$$\begin{aligned} \bar{\ell}_t(\bar{x}_t, \bar{u}_t) + \kappa\,\theta(1-\theta)\gamma(\|(x_t - x_t^\star, u_t - u_t^\star)\|) &\leqslant \theta\bar{\ell}_t(x_t, u_t) + (1-\theta)\bar{\ell}_t(x_t^\star, u_t^\star) = \theta\bar{\ell}_t(x_t, u_t) \\ \kappa\,(1-\theta)\gamma(\|(x_t - x_t^\star, u_t - u_t^\star)\|) &< \bar{\ell}_t(x_t, u_t) \end{aligned} \tag{3.35}$$

which implies strict dissipativity as defined in 3.8 with $\alpha(r) := \kappa\,(1-\theta)\gamma(r) \in \mathcal{K}_\infty$. □

Theorem 3.5 not only establishes strict dissipativity but also guarantees the continuity of the storage function $\lambda_t(x_t, u_t)$.

Lemma 3.6. *Given that we achieve strict dissipativity through Theorem 3.5 our storage function $\lambda_t(x_t, u_t)$ is continuous with $\gamma_\lambda(\|x_t - x_t^\star\|) \in \mathcal{K}_\infty$ such that*

$$|\lambda_t(x) - \lambda_t(x_t^\star)| \leqslant \gamma_\lambda(\|x_t - x_t^\star\|) \tag{3.36}$$

for all $x \in \mathcal{X}$.

Proof. According to Theorem 3.5 our storage function is given by Equation (3.27) which gives us

$$|\lambda_t(x) - \lambda_t(x_t^\star)| = \left|\frac{q_t^\top}{\eta_0}(x_t - x_t^\star)\right| \leqslant \frac{\tilde{q}^\top}{\eta_0}\|x_t - x_t^\star\| \tag{3.37}$$

with $\tilde{q} = \max_{t\in\mathbb{N}} |q_t|$. Since the sequence q_t is bounded, as shown in the proof of Theorem [63, Theorem 4.1], $\tilde{q}$ is also bounded, and by Theorem 3.5 we further have $\eta_0 > 0$. □

The following property, which is taken from [49, Assumption 1] and referred to as cheap reachability, is needed in order to show Lemma 3.8 which gives us the Turnpike property.

Assumption 3.7. [Cheap Reachability] The overtaking optimal pair $(x^\star, u^\star)$ is said to be *cheaply reachable* if there exist constants $E, \tilde{E} \in \mathbb{R}$ such that, for all $t, N \in \mathbb{N}$ and $x(t) \in \mathcal{X}$, the following inequalities are satisfied

$$\hat{V}_N(x(t), t) \leqslant E, \quad \tilde{V}_N(x(t), t) \leqslant \tilde{E}. \tag{3.38}$$

This assumption requires that it is possible to reach the optimal overtaking trajectory in finite time from any initial state within $\mathcal{X}$, ensuring a bounded overall cost. This boundedness arises because the shifted cost along the optimal trajectory $x^\star$ is zero. A stronger, more restrictive assumption that guarantees such reachability is n_x-step reachability (cf. [66]), where any state can transition to the optimal trajectory within n_x steps or fewer.

Given our previous assumptions, we can now show Turnpike for both our rotated and regular OCP as required in [13, Assumptions 1, 2 and 5]. This condition is later used in [13, Theorem 4] to show convergence to the infinite-horizon optimal trajectory.

Lemma 3.8. *Given that Assumption 3.2-3.7 and the one of Theorem 3.5 hold, the OCP is strictly dissipative and cheaply reachable with respect to the overtaking optimal trajectory $(x^\star, u^\star)$. As a result, the Turnpike property from Definition 3.10 holds for both the modified and regular OCP, in both finite and infinite horizon.*

Proof. By applying Assumption 3.2 along with the conditions from Theorem 3.5, we can use Theorem 3.5 to establish strict dissipativity. With strict dissipativity and Assumption 3.7, we then apply [49, Theorem 1] to conclude the Turnpike property as defined in Definition 3.10. □

The following two assumptions are needed in order to establish continuity of the optimal value functions as defined in [13, Assumption 3 and 7] which is required for showing convergence to the infinite-horizon optimal trajectory according to [13, Theorem 4].

Assumption 3.9. [Local Controllability] Assume that the system is *locally controllable* along the trajectory pair $(x^\star, u^\star)$, i.e. there exists a time $\tau \in \mathbb{N}, \delta_c > 0$, and $\gamma_x(\delta), \gamma_u(\delta), \gamma_c(\delta) \in \mathcal{K}_\infty$ such that for each $t \in \mathbb{N}$ and for any two points $x_t \in \mathcal{B}_{\delta_c}(x_t^\star)$, $y_{t+\tau} \in \mathcal{B}_{\delta_c}(x_{t+\tau}^\star)$ in a δ_c-neighborhood around the optimal trajectory, there exists an input $u_{t+k} \in \mathcal{U}$ such that the trajectory starting at x_t satisfies $x_{t+\tau} = y_{t+\tau}$ and the estimates

$$\|x_{t+k} - x_{t+k}^\star\| \leqslant \gamma_x(\delta) \tag{3.39}$$

$$\|u_{t+k} - u_{t+k}^\star\| \leqslant \gamma_u(\delta) \tag{3.40}$$

$$|\hat{\ell}_t(x_{t+k}, u_{t+k})| \leqslant \gamma_c(\delta) \tag{3.41}$$

for all $k \in \mathbb{N}_\tau$, where $\delta := \max\{\|x_t - x_t^\star\|, \|y_{t+\tau} - x_{t+\tau}^\star\|\}$.

This assumption essentially requires that any two points $(x_t, y_{t+\tau})$ within a δ_c-vicinity of the optimal trajectory can be connected by a sub-optimal trajectory in forward time. Figure 3.3 illustrates such a trajectory, starting from x_t reaching $y_{t+\tau}$ while remaining within the δ_c-neighborhood of the optimal trajectory.

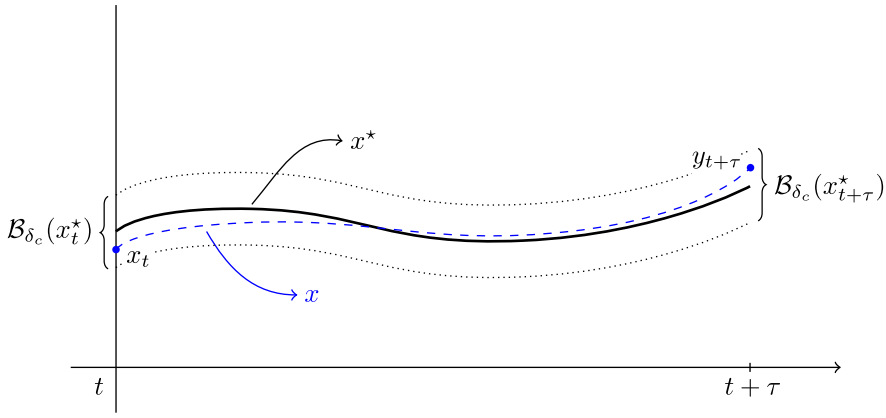

Figure 3.3: Local controllability of trajectory x around the optimal trajectory $x^\star$ (Figure adapted from [54, Figure 5.1]).

Assumption 3.10. [Continuity and Upper Bound] There exist two functions $\gamma_\ell(\|(x_1-x_2, u_1-u_2)\|)$ and $\gamma_{\tilde{\ell}}(\|(x_t - x_t^\star, u_t - u_t^\star)\|) \in \mathcal{K}_\infty$ such that the stage cost $\ell_t(x_t, u_t)$ and the rotated stage cost $\tilde{\ell}_t(x_t, u_t)$ from Definition 3.9 satisfy the inequality

$$|\ell_t(x_1, u_1) - \ell_t(x_2, u_2)| \leqslant \gamma_\ell(\|(x_1 - x_2, u_1 - u_2)\|) \tag{3.42}$$

$$\tilde{\ell}_t(x_t, u_t) \leqslant \gamma_{\tilde{\ell}}(\|(x_t - x_t^\star, u_t - u_t^\star)\|) \tag{3.43}$$

for all $(x_t, u_t) \in \mathcal{X} \times \mathcal{U}$ and $(x_1, u_1), (x_2, u_2) \in \mathcal{Y}$ that lie in the compact subset $\mathcal{Y} \subseteq \mathcal{X} \times \mathcal{U}$. The former inequality can be understood as a form of local Lipschitz-continuity while the latter one provides an upper-bound on the rotated stage cost similar to the lower-bound provided by strict dissipativity.

Given these assumptions we can finally present following Lemma which gives us continuity for both the rotated and regular value function.

Lemma 3.11. *The value function of problem* (3.12) *subject to the constraints defined in* (3.24) *is continuous according to [13, Assumption 3], and its rotated value function is continuous, as required in [13, Assumption 7], provided that Assumptions 3.2-3.10 are satisfied.*

Proof. We first establish continuity of the regular value function. To that end notice that under Assumption 3.2 and the conditions of 3.5, strict dissipativity holds. By applying [49, Theorem 2] with Assumption 3.7-3.10, we obtain an (approximately) continuous value function as defined in [49, Definition 6].

Let us now analyze the rotated value function of problem (3.12). Given the strict dissipativity of our OCP, strict dissipativity must also hold for the rotated OCP with the storage function $\tilde{\lambda}_t(x_t) = 0$. Given Assumption 3.7-3.10 we can again apply [49, Theorem 2] which gives us an (approximately) continuous rotated value function as defined below

$$|\tilde{V}_N(x_t, t) - \tilde{V}_N(x_t^\star, t)| \leqslant \gamma_{\tilde{V}}(N, \|x_t - x_t^\star\|) \tag{3.44}$$

with $\gamma_{\tilde{V}}$ being monotonically decreasing in N. By applying the following worst-case assumption we obtain the upper bound required by [13, Assumption 7] given by

$$|\tilde{V}_N(x_t, t) - \tilde{V}_N(x_t^\star, t)| \leqslant \tilde{\gamma}_{\tilde{V}}(\|x_t - x_t^\star\|). \tag{3.45}$$

where $\tilde{\gamma}_{\tilde{V}}(\|x_t - x_t^\star\|) := \gamma_{\tilde{V}}(1, \|x_t - x_t^\star\|)$. □

We now summarize the key result from [13, Theorem 1-4], that follow directly from Assumption 3.2-3.10 and Lemma 3.4-3.11.

(1) [13, Theorem 1] ensures that the closed-loop trajectory approaches the optimal trajectory in an approximately optimal way.

(2) [13, Theorem 2] provides $\mathbb{P}$-practical asymptotic stability given that there a exists a uniform time-varying Lyapunov function.

(3) [13, Theorem 4] establishes the existence of a Lyapunov function for the closed-loop system together with a $\mathbb{P}$-practical set that shrinks to zero as the optimization horizon N tends to infinity.

The following figure provides a visual summary of these theoretical results, illustrating how they relate to the closed-loop system's behavior and stability properties.

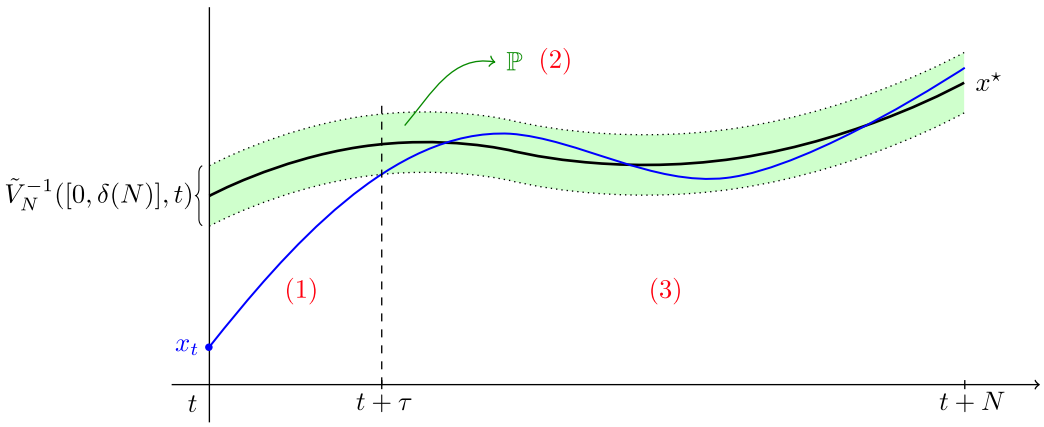

Figure 3.4: The closed-loop state trajectory x (blue) approaches the optimal trajectory $x^\star$ (black) in an approximately optimal way within τ time steps, eventually converging to the $\mathbb{P}$-practical asymptotic stability set (green shaded). Hereby the invariant set $\mathbb{P}$ is given by the inverse of the rotated optimal value function $\tilde{V}_N([0,\delta(N)],t)$ which is approaching zero with increasing horizons N since $\delta(N) \in \mathcal{L}$.

The proofs of the theorems referenced can be found in [13]. Specifically, the proofs of the theorems required to establish Lemma 3.8 and Lemma 3.11 are provided in [49]. Additionally, illustrative figures explaining the concepts discussed in these two papers can be found in Pirkelmann's dissertation [54], which also contains both papers.

Recursive Feasibility

Recursive feasibility can be established based on the previously stated assumptions and lemmas. Specifically, given Assumptions 3.2-3.7 and the conditions of Theorem 3.5, one can invoke Lemma 3.8, which establishes the Turnpike property. Combined with Assumption 3.9, this ensures recursive feasibility as stated in [53, Proposition 4.2]. While the proof in [53, Proposition 4.2] was originally formulated for time-invariant systems based on local n_x-step reachability, it extends to time-varying systems with only minor modifications by replacing local n_x-step reachability with the local controllability condition in Assumption 3.9.

3.2.3 Economic MPC with Terminal Ingredients

In contrast to EMPC without terminal ingredients, a substantial body of literature has been developed on EMPC with terminal ingredients, including approaches that utilize terminal constraints

and terminal costs to ensure stability and recursive feasibility. For instance, in [41], asymptotic stability and recursive feasibility are established for discrete-time LTI systems with a time-varying stage cost, under the condition that the stage cost remains constant over the prediction horizon of the MPC controller. In this approach, a precomputed optimal steady state and its corresponding terminal set are used to guarantee recursive feasibility and ensure stability. Naturally, this requires recalculating the optimal steady state whenever the stage cost changes. The work of [42] establishes closed-loop stability and recursive feasibility for continuous time, time-invariant, nonlinear dynamics together with a time-varying objective based on a Lyapunov controller which guarantees save operation. The drawback of this method is that one needs to find a Lyapunov controller that provides a control-invariant set for different operating points which is not necessarily trivial and can potentially lead to an overly conservative region of attraction. [67] compares and summarizes different discrete time EMPC methods giving only asymptotic performance bounds, highlighting the difficulty of deriving asymptotic stability certificates for time-varying cost functions. The paper [44] presents a method capable of dealing with periodic optimal control for general nonlinear discrete-time systems and periodically time-varying stage cost providing recursive feasibility, constraint satisfaction and performance guarantees. This is achieved with an artificial reference trajectory that converges to the locally optimal periodic orbit, ensuring that the closed-loop average performance is no worse than that of the potentially limiting periodic orbit. Accordingly it follows that the average performance of the closed-loop system is no worse than the globally optimal periodic orbit for linear systems with convex constraints and cost. The authors from [68] propose an EMPC framework that guarantees asymptotic stability and recursive feasibility for general continuous time, time-varying systems and cost, relying on an auxiliary control law that asymptotically drives the trajectory to the average economically optimal trajectory pair. It provides two constructive methods on how to construct these auxiliary control laws without the need to explicitly compute the storage function of the stage cost.

We establish asymptotic stability and recursive feasibility using terminal ingredients based on the work of [14] which allows general time-varying, nonlinear dynamics and a general time-varying objective with just one terminal constraint and no terminal cost. This is very desirable for our use case as every additional terminal constraint does increase complexity of the game, i.e., the complexity for individual agents. To this end, we consider the general dynamics defined in (3.9), subject to the closed state constraint set $\mathcal{X}$ and the compact input constraint set $\mathcal{U}$. Let the pair (x^r, u^r) represent an arbitrary reference trajectory for the nonlinear system, satisfying the dynamics for all $t \in \mathbb{N}$

$$x^r_{t+1} = f_t(x^r_t, u^r_t), \tag{3.46}$$

where $(x^r_t, u^r_t) \in \mathcal{X} \times \mathcal{U}$.

Given the above defined reference trajectory x^r_t we now define our OCP with terminal constraint

$$\begin{aligned} V_N(x(t), t) := \min_{x_{\cdot|t}, u_{\cdot|t}} \quad & J_N\big(x_{\cdot|t}, u_{\cdot|t}, t\big) \\ \text{s.t.} \quad & x_{0|t} = x(t), \\ & x_{k+1|t} = f_{t+k}(x_{k|t}, u_{k|t}), && \forall\, k \in \mathbb{N}_N \\ & x_{k|t} \in \mathcal{X}, && \forall\, k \in \mathbb{N}_{N+1} \\ & u_{k|t} \in \mathcal{U}, && \forall\, k \in \mathbb{N}_N \\ & x_{N|t} = x^r_{t+N}. \end{aligned} \tag{3.47}$$

Stability

In this section we will establish the assumption necessary to apply the key results from [14] essentially providing asymptotic stability and an average performance bound in case the system is not dissipative with respect to the reference trajectory x^r (see Definition 3.8).

We star with some mild assumption on our dynamics and stage cost.

Assumption 3.12. [Continuity Conditions] The dynamics $f_t(x_t, u_t)$ is continuous and the stage cost $\ell_t(x_t, u_t)$ is lower semicontinuous for all $t \in \mathbb{N}$.

The next assumption imposes an additional condition on the stage cost, needed to give asymptotic average performance bounds.

Assumption 3.13. [Bounded Below] The stage cost $\ell_t(x_t, u_t)$ is uniformly bounded below for $(x_t, u_t) \in \mathcal{X} \times \mathcal{U}$, for all $t \in \mathbb{N}$.

Given those two assumptions we can now state [14, Theorem 1] which gives us a performance bound for the asymptotic average performance of the MPC (3.47) with respect to the reference trajectory defined in (3.46).

Theorem 3.14. *Under Assumption 3.12 and 3.13 and given a feasible initial state $x(t)$ the following average performance bound holds for all $t \in \mathbb{N}$*

$$\limsup_{N\to\infty} \frac{1}{N} \sum_{k=t}^{t+N-1} \ell_t(x_k, u_k) - \ell_t(x_t^r, u_t^r) \leqslant 0 \tag{3.48}$$

In other words, the closed-loop trajectory obtained from successively applying the first input of (3.47) results in an asymptotic average cost that is at least as good as the one of our potentially limiting reference trajectory.

Additionally imposing the following two assumptions provides us with asymptotic stability guarantees with respect to our reference trajectory (x^r, u^r).

Assumption 3.15. [Strict Dissipativity of the Reference Trajectory] The system (3.47) is strictly dissipative with respect to the reference trajectory (x^r, u^r), as defined in (3.46), and the supply rate $s_t(x_t, u_t) = \ell_t(x_t, u_t) - \ell_t(x_t^r, u_t^r)$ in accordance with Definition 3.8. Specifically, there exists a storage function $\lambda_t(x_t)$ such that $\lambda_t(x_t^r) = 0$.

We further require the following continuity assumption to hold.

Assumption 3.16. [Continuous Rotated Value Function] The rotated optimal value function[3] $\tilde{V}_N(x_t, t)$ is uniformly continuous at (x_t^r) such that there exists an upper bound $\gamma(\|x_t - x_t^r\|) \in \mathcal{K}_\infty$ that

$$\tilde{V}_N(x_t, t) \leqslant \gamma(\|x_t - x_t^r\|) \tag{3.49}$$

holds for all $t \in \mathbb{N}$.

Remark 8. Note that this condition is equivalent to the one proposed by Lemma 3.11 Equation (3.45) as one can reformulate (3.49) as follows

$$\tilde{V}_{N,t}(x_t) = \tilde{V}_N(x_t, t) - \tilde{V}_N(x_t^r, t) \leqslant |\tilde{V}_N(x_t, t) - \tilde{V}_N(x_t^r, t)| \leqslant \gamma(\|x_t - x_t^r\|) \tag{3.50}$$

taking into account that $\tilde{V}_N(x_t^r, t) = 0$ for all $t \in \mathbb{N}$ as $\lambda_t(x_t^r) = 0$.

Based on these assumptions we can now present the main finding from [14, Theorem 2] providing asymptotic stability of our EMPC.

Theorem 3.17. *Let Assumptions 3.12, 3.15, 3.16 hold. Then, there exists a $\mathcal{KL}$ function $\beta(|x_t - x_t^r|, k - t)$ such that for any feasible initial state $x(t)$ with $t \in \mathbb{N}$*

$$\|x_{t+k} - x_{t+k}^r\| \leqslant \beta(\|x_{t+k} - x_{t+k}^r\|, k - t) \tag{3.51}$$

holds for all times $k \geqslant t$.

[3] In [14], the rotated optimal value function $\tilde{V}_N(x_t, t)$ is defined slightly differently. However, since we enforce an exact terminal constraint $x_{N|t} = x_{t+N}^r$ with $\lambda_t(x_t^r) = 0$, its definition aligns with the notation used here.

Recursive Feasibility

Recursive feasibility is directly obtained from the terminal constraint x_t^r which allows to simply shift the solution to the OCP by one time-step as x_t^r is a recursively feasible trajectory itself. Hence a feasible input is always given by the shifted suboptimal input given by $\tilde{u}_{\bullet|t+1} = \left\{u_{1|t}, \ldots, u_{N-1|t}, u_{t+N}^r\right\}$.

Chapter 4

Time-Varying Receding Horizon Games

In this chapter we first formally introduce receding horizon games and show how to connect them to EMPC theory under the assumption of a potential game structure. In an next step we analyze the class of linear quadratic games in the context of RHGs. This type of game imposes a very specific form of cost function though still allows to capture a wide range of different system behaviors as visible from Table 3.1. Further, we are going to establish the basic conditions on our cost such that it falls into the class of PGs, hence possesses a potential function. From there we will demonstrate asymptotic stability once without terminal conditions, i.e., the game as is, and once with terminal conditions in form of a reference trajectory. Finally we are going to compare the performance of both methods in simulation based on the example of demand-side management (DSM) provided by [7].

4.1 Receding Horizon Games

In this section, we formally introduce RHGs and relate them to EMPC theory. Lets consider an agent's discrete-time, time-varying, nonlinear dynamics governed by

$$x_{t+1}^v = f_t^v(x_t^v, u_t^v) \tag{4.1}$$

subject to the hard state and input constraints

$$x_t^v \in \mathcal{X}^v\big(\boldsymbol{x}_t^{-v}\big) := \big\{x_t^v \in \mathbb{R}^{n^v} \mid (x_t^v, \boldsymbol{x}_t^{-v}) \in \boldsymbol{\mathcal{X}}\big\} \tag{4.2}$$

$$u_t^v \in \mathcal{U}^v\big(\boldsymbol{u}_t^{-v}\big) := \big\{u_t^v \in \mathbb{R}^{m^v} \mid (u_t^v, \boldsymbol{u}_t^{-v}) \in \boldsymbol{\mathcal{U}}\big\} \tag{4.3}$$

with $\boldsymbol{\mathcal{X}}$ and $\boldsymbol{\mathcal{U}}$ denoting the time-invariant coupling constraints of all agents' decision variables. This structure of coupling constraints is referred to a jointly convex constraints, as per [16, Definition 2] We require such a structure for the following results.

The RHG is then given by the following coupled optimization problem where each agent v aims to find a strategy $(x_{\cdot|t}^v, u_{\cdot|t}^v)$ that minimizes its own cost $J_N^v(\boldsymbol{x}_{\cdot|t}, \boldsymbol{u}_{\cdot|t}, t)$ given the other agent's

strategies $(\boldsymbol{x}_{\bullet|t}^{-v}, \boldsymbol{u}_{\bullet|t}^{-v})$

$$\forall\, v \in \mathcal{A}: \quad \begin{cases} V_N^v(x^v(t), t) := \min\limits_{x_{\bullet|t}^v, u_{\bullet|t}^v} & J_N^v\big(\boldsymbol{x}_{\bullet|t}, \boldsymbol{u}_{\bullet|t}, t\big) \\ \qquad\qquad \text{s.t.} & x_{0|t}^v = x^v(t), \\ & x_{k+1|t}^v = f_{t+k}^v(x_{k|t}^v, u_{k|t}^v), \quad \forall\, k \in \mathbb{N}_N \\ & x_{k|t}^v \in \mathcal{X}^v\big(\boldsymbol{x}_{k|t}^{-v}\big), \quad \forall\, k \in \mathbb{N}_{N+1} \\ & u_{k|t}^v \in \mathcal{U}^v\big(\boldsymbol{u}_{k|t}^{-v}\big) \quad \forall\, k \in \mathbb{N}_N \end{cases} \tag{4.4}$$

with $x^v(t)$ being the initial state of agent v at time instance t and the optimal value function $V_N^v(x^v(t), t)$. Furthermore, the cost of agent v is defined as

$$J_N^v\big(\boldsymbol{x}_{\bullet|t}, \boldsymbol{u}_{\bullet|t}, t\big) := \sum_{k=0}^{N-1} \ell_{t+k}^v(\boldsymbol{x}_{k|t}, \boldsymbol{u}_{k|t}) \tag{4.5}$$

with the time-varying stage cost $\ell_t^v(\boldsymbol{x}_t, \boldsymbol{u}_t)$.

Remark 9. In Section 4.2 we alternatively denote the stage cost by $\ell^v(\boldsymbol{x}_t, \boldsymbol{u}_t, \Theta_t^v)$ to emphasize that the general structure of the cost remains fixed while only the time-varying parameter Θ_t^v changes.

In the following, we choose the GNE as defined in (3.4) as a solution to this coupled OCP. We will later establish the conditions for which this solution exists.

4.1.1 Connecting Receding Horizon Games and EMPC

Our general goal is to provide stability and recursive feasibility guarantees for the GNEP defined in (4.4). We achieve that for a special class of games. i.e., potential games which allow us to reformulate our GNEP into a centralized OCP as follows

$$\begin{aligned} V_N(\boldsymbol{x}(t), t) := \min_{\boldsymbol{x}_{\bullet|t}, \boldsymbol{u}_{\bullet|t}} \quad & \boldsymbol{p}_N\big(\boldsymbol{x}_{\bullet|t}, \boldsymbol{u}_{\bullet|t}, t\big) \\ \text{s.t.} \quad & \boldsymbol{x}_{0|t} = \boldsymbol{x}(t), \\ & \boldsymbol{x}_{k+1|t} = \boldsymbol{f}_{t+k}(\boldsymbol{x}_{k|t}, \boldsymbol{u}_{k|t}), \quad \forall\, k \in \mathbb{N}_N \\ & \boldsymbol{x}_{k|t} \in \boldsymbol{\mathcal{X}}, \quad \forall\, k \in \mathbb{N}_{N+1} \\ & \boldsymbol{u}_{k|t} \in \boldsymbol{\mathcal{U}} \quad \forall\, k \in \mathbb{N}_N \end{aligned} \tag{4.6}$$

where we denote the stacked initial state of all agents with $\boldsymbol{x}(t)$, the stacked dynamics using $\boldsymbol{f}_t(\boldsymbol{x}_{k|t}, \boldsymbol{u}_{k|t})$ and the potential function $\boldsymbol{p}_N\big(\boldsymbol{x}_{\bullet|t}, \boldsymbol{u}_{\bullet|t}, t\big)$ as the cost of the centralized OCP (see Equation (3.7)). This centralized receding horizon game can then be analyzed using tools from economic model predictive control theory.

4.2 The Potential Function

In this section, we analyze linear quadratic games within the RHG framework defined in (4.4). We establish conditions under which the game exhibits a potential game structure and derive a corresponding potential function. To this end, we consider a time-varying stage cost composed of linear and quadratic terms

$$\ell(\boldsymbol{x}_t, \boldsymbol{u}_t, \Theta_t^v) = \boldsymbol{x}_t^\top \boldsymbol{Q}_t^v \boldsymbol{x}_t + \boldsymbol{u}_t^\top \boldsymbol{R}_t^v \boldsymbol{u}_t + s_t^v x_t^v + c_t^v u_t^v. \tag{4.7}$$

where $\boldsymbol{Q}_t^v \in \mathbb{R}^{n\times n}$ denotes the sparse block matrix with nonzero entries only in the v-th block row

$$\boldsymbol{Q}_t^v := \begin{bmatrix} & & & \\ Q_t^{v,1} & \cdots & Q_t^{v,v} & \cdots & Q_t^{v,M} \\ & & & \end{bmatrix} \tag{4.8}$$

and $Q_t^{i,j} \in \mathbb{R}^{n^i\times n^j}$. $\boldsymbol{R}_t^v$ is defined analogously to $\boldsymbol{Q}_t^v$ and s_t^v as well as c_t^v are row-vectors of compatible dimension.

Remark 10. The reason $\boldsymbol{Q}_t^v$ has only nonzero entries in the v-th block row is that all entries outside the v-th block row and column depend solely on the decision variables of other agents and therefore, do not influence the OCP of agent v. Consequently, these terms can be set to zero, as they are constant offsets that have no impact on the optimization.

Similar to Equation (4.8), one could alternatively place nonzero entries in the v-th block column of $\boldsymbol{Q}_t^v$, leading to the matrix defined as follows

$$\boldsymbol{Q}_t^v := \begin{bmatrix} & \tilde{Q}_t^{1,v} & \\ & \vdots & \\ \tilde{Q}_t^{v,1} \cdots & \tilde{Q}_t^{v,v} & \cdots \tilde{Q}_t^{v,M} \\ & \vdots & \\ & \tilde{Q}_t^{M,v} & \end{bmatrix}. \tag{4.9}$$

However, this formulation can ultimately be reduced back to the structure in Equation (4.8), which can be seen by evaluating the quadratic form

$$\boldsymbol{x}_t^\top \boldsymbol{Q}_t^v \boldsymbol{x}_t = \sum_{i=1}^{M} (x_t^v)^\top \tilde{Q}_t^{v,i}\, x_t^i + (x_t^i)^\top \tilde{Q}_t^{i,v}\, x_t^v = \sum_{i=1}^{M} (x_t^v)^\top \underbrace{\left((\tilde{Q}_t^{i,v})^\top + \tilde{Q}_t^{v,i}\right)}_{\Leftrightarrow\, Q_t^{v,i}} x_t^i = \sum_{i=1}^{M} (x_t^v)^\top Q_t^{v,i}\, x_t^i. \tag{4.10}$$

Given this sparse problem structure, we can alternatively rewrite our stage cost as the following summation

$$\ell(\boldsymbol{x}_t, \boldsymbol{u}_t, \Theta_t^v) = \sum_{i=1}^{M} \left((x_t^v)^\top Q_t^{v,i}\, x_t^i + (u_t^v)^\top R_t^{v,i}\, u_t^i \right) + s_t^v x_t^v + c_t^v u_t^v. \tag{4.11}$$

In order to ensure that OCP (3.12) with stage cost (4.8) satisfies the properties of a PG we make the following assumption on the off-diagonal terms of $\boldsymbol{Q}_t^v$ and $\boldsymbol{R}_t^v$.

Assumption 4.1. [Symmetric off-diagonal Terms] The off-diagonal terms of the sparse block cost matrices are symmetric

$$Q_t^{i,j} = (Q_t^{j,i})^\top, \quad R_t^{i,j} = (R_t^{i,j})^\top \qquad \forall\, i \neq j \in \mathcal{A}. \tag{4.12}$$

Under Assumption 4.1, the exact potential function for this problem is given by the following theorem.

Theorem 4.2. *Given that Assumption 4.1 holds the exact potential function for this problem is given by*

$$\boldsymbol{p}_N(\boldsymbol{x}, \boldsymbol{u}, t) = \sum_{k=t}^{t+N-1} \boldsymbol{p}(\boldsymbol{x}_k, \boldsymbol{u}_k, \Theta_k) \tag{4.13}$$

with the stage potential function defined as

$$\begin{aligned}\boldsymbol{p}(\boldsymbol{x}_t, \boldsymbol{u}_t, \Theta_t) = \sum_{i=1}^{M} s_t^i x_t^i + c_t^i u_t^i \\ + \frac{1}{2}\sum_{i=1}^{M}\sum_{j=1}^{M}\left((x_t^i)^\top Q_t^{i,j}\, x_t^j + (u_t^i)^\top R_t^{i,j}\, u_t^j\right) \\ + \frac{1}{2}\sum_{i=1}^{M}(x_t^i)^\top Q_t^{i,i}\, x_t^i + (u_t^i)^\top R_t^{i,i}\, u_t^i. \end{aligned} \tag{4.14}$$

This can alternatively be written in compact matrix representation with the dense block matrices

$$\boldsymbol{Q}_t := \frac{1}{2}\begin{bmatrix} 2Q_t^{1,1} & Q_t^{1,2} & \dots & Q_t^{1,M} \\ Q_t^{2,1} & 2Q_t^{2,2} & \dots & Q_t^{2,M} \\ \vdots & \ddots & \ddots & \vdots \\ Q_t^{M,1} & \dots & \dots & 2Q_t^{M,M} \end{bmatrix}, \quad \boldsymbol{R}_t := \frac{1}{2}\begin{bmatrix} 2R_t^{1,1} & R_t^{1,2} & \dots & R_t^{1,M} \\ R_t^{2,1} & 2R_t^{2,2} & \dots & R_t^{2,M} \\ \vdots & \ddots & \ddots & \vdots \\ R_t^{M,1} & \dots & \dots & 2R_t^{M,M} \end{bmatrix} \tag{4.15}$$

and the stacked vectors $\boldsymbol{s}_t = \left[s_t^1, \dots, s_t^M\right]$ *and* $\boldsymbol{c}_t = \left[c_t^1, \dots, c_t^M\right]$ *such that*

$$\boldsymbol{p}(\boldsymbol{x}_t, \boldsymbol{u}_t, \Theta_t) = \boldsymbol{x}_t^\top \boldsymbol{Q}_t \boldsymbol{x}_t + \boldsymbol{u}_t^\top \boldsymbol{R}_t \boldsymbol{u}_t + \boldsymbol{s}_t \boldsymbol{x}_t + \boldsymbol{c}_t \boldsymbol{u}_t. \tag{4.16}$$

Proof. It can immediately be seen from [17, Theorem 4.5] that Assumption 4.1 must hold in order for problem (4.7) to be an exact PG. However, we now proceed with a constructive proof. Consider the gradients of the agents' stage costs with respect to x_t^v and u_t^v

$$\frac{\partial\, \ell(\boldsymbol{x}_t, \boldsymbol{u}_t, \Theta_t^v)}{\partial\, x_t^v} = (s_t^v)^\top + \left(Q_t^{v,v} + (Q_t^{v,v})^\top\right) x_t^v + \sum_{i \in \mathcal{A}\setminus v} Q_t^{v,i}\, x_t^i \tag{4.17}$$

$$\frac{\partial\, \ell(\boldsymbol{x}_t, \boldsymbol{u}_t, \Theta_t^v)}{\partial\, u_t^v} = (c_t^v)^\top + \left(R_t^{v,v} + (R_t^{v,v})^\top\right) u_t^v + \sum_{i \in \mathcal{A}\setminus v} R_t^{v,i}\, u_t^i \tag{4.18}$$

together with the gradients of the stage potential function stated in Equation 4.14

$$\frac{\partial\, \boldsymbol{p}(\boldsymbol{x}_t, \boldsymbol{u}_t, \Theta_t)}{\partial\, x_t^v} = (s_t^v)^\top + \left(Q_t^{v,v} + (Q_t^{v,v})^\top\right) x_t^v + \frac{1}{2}\sum_{i \in \mathcal{A}\setminus v} \left(Q_t^{v,i} + (Q_t^{i,v})^\top\right) x_t^i \tag{4.19}$$

$$\frac{\partial\, \boldsymbol{p}(\boldsymbol{x}_t, \boldsymbol{u}_t, \Theta_t)}{\partial\, u_t^v} = (c_t^v)^\top + \left(R_t^{v,v} + (R_t^{v,v})^\top\right) u_t^v + \frac{1}{2}\sum_{i \in \mathcal{A}\setminus v} \left(R_t^{v,i} + (R_t^{i,v})^\top\right) u_t^i. \tag{4.20}$$

According to Lemma 3.1 the gradients of the cost and the potential function must be equal for all $v \in \mathcal{A}$ in order for $\boldsymbol{p}_N(\boldsymbol{x}, \boldsymbol{u}, t)$ to be an exact potential function. It can be seen that Assumption 4.1 guarantees that the gradient of the stage cost equals the gradient of the stage potential function for all t such that $\boldsymbol{p}_N(\boldsymbol{x}, \boldsymbol{u}, t)$ is an exact potential function. □

Remark 11. The linear contribution to the stage cost $s_t^v x_t^v + c_t^v u_t^v$ can easily be generalized by replacing it with an arbitrary, continuous, differentiable cost function depending only on agent v such as $q_t^v(x_t^v, u_t^v)$ (see, e.g., [17, Equation 1.2]).

Remark 12. For the special case of

$$\forall\, i, j \in \mathcal{A}: \quad \begin{cases} Q_t^{i,i} = Q_t^{i,j} = \frac{1}{M}\mathcal{I}_{n^v} \\ R_t^{i,i} = R_t^{i,j} = \frac{1}{M}\mathcal{I}_{n^m} \end{cases} \tag{4.21}$$

we recover an aggregative game as, for example, defined in [5].

4.3 Stability without Terminal Ingredients

With the previously derived potential function we can now reformulate our GNEP into a centralized OCP (4.6) as described in Section 4.1.1 such that we can show stability using the tools from EMPC theory outlined in Section 3.2. More specifically, to guarantee stability without terminal conditions, we rely on the results from Section 3.2.2. Hereby we consider LTV dynamics as defined in (3.23), denoting the stacked dynamics of all agents v as

$$\boldsymbol{x}_{t+1} = \boldsymbol{A}_t \boldsymbol{x}_t + \boldsymbol{B}_t \boldsymbol{u}_t + \boldsymbol{d}_t \tag{4.22}$$

with $\boldsymbol{A}_t$ the stacked state transition matrix, $\boldsymbol{B}_t$ the stacked input matrix and $\boldsymbol{d}_t$ the stacked affine offset. We further assume that the state constraint set $\boldsymbol{\mathcal{X}}$ is a nonempty, bounded, open subset of $\mathbb{R}^n$ and that the system is subject to the convex input constraints $\boldsymbol{\mathcal{U}}$ defined in (3.24). These assumptions are required by Theorem 3.3 Condition (1), which is needed to establish dissipativity.

As mentioned already in Section 3.2.2 we assume that $\boldsymbol{A}_t$ is full rank for all $t \in \mathbb{N}_N$. This is normally given for discretized systems obtained from continuous time systems as discretization methods, such as Euler or Runge-Kutta-4 (RK4), normally produce matrices of full rank.

We take the following three assumption starting with a condition on the cost as defined below.

Assumption 4.3. [Assumptions on Cost] There exist constants $\underline{\boldsymbol{\sigma}}_x, \overline{\boldsymbol{\sigma}}_x, \underline{\boldsymbol{\sigma}}_u, \overline{\boldsymbol{\sigma}}_u > 0$ such that the quadratic cost matrices are positive definite and bounded for all $t \in \mathbb{N}_N$ according to

$$\underline{\boldsymbol{\sigma}}_x \mathcal{I} \preceq \boldsymbol{Q}_t \preceq \overline{\boldsymbol{\sigma}}_x \mathcal{I}, \quad \underline{\boldsymbol{\sigma}}_u \mathcal{I} \preceq \boldsymbol{R}_t \preceq \overline{\boldsymbol{\sigma}}_u \mathcal{I}. \tag{4.23}$$

We also assume that the linear cost terms are bounded

$$\|\boldsymbol{s}_t\| \leqslant \overline{\boldsymbol{s}}, \quad \|\boldsymbol{c}_t\| \leqslant \overline{\boldsymbol{c}} \tag{4.24}$$

with the constants $\overline{\boldsymbol{s}}, \overline{\boldsymbol{c}} \geqslant 0$ for all $t \in \mathbb{N}_N$.

In addition to the assumption on the cost we require n_x-step reachability to ensure boundedness of the optimal value functions as required by Assumption 3.7 [66, Definition 1].

Assumption 4.4. [n_x-step Reachability] We assume that the system defined in (4.22) is *n_x-step reachable* in the sense that there exists a finite $n_x \in \mathbb{N}$ for every state $\boldsymbol{x}' \in \boldsymbol{\mathcal{X}}$, such that $\boldsymbol{x}'$ can be reached from the origin $\boldsymbol{x}_0 = 0$ in n_x-steps, i.e., $\boldsymbol{x}_{n_x} = \boldsymbol{x}'$, using an admissible control sequence $\boldsymbol{u}_t \in \boldsymbol{\mathcal{U}}$ for all $t \in \mathbb{N}_{n_x}$.

To make sure that the system remains controllable close to the optimal trajectory as well, we further assume local controllability as defined in Assumption 3.9.

Given Assumption 4.3 we fulfill uniform strict convexity (Assumption 3.2) and continuity of the stage potential function (3.42) together with upper boundedness of the rotated stage potential function as defined in (3.43).

Lemma 4.5. *Consider the system defined in* (4.22) *subject to the constraints $\boldsymbol{\mathcal{X}} \times \boldsymbol{\mathcal{U}}$ and let assumptions 4.3 hold. Then Assumptions 3.2 and 3.10 are satisfied, providing uniform strict convexity and continuity of the stage potential function $\boldsymbol{p}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$, together with an upper bound to the rotated stage potential function $\tilde{\boldsymbol{p}}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$.*

Proof. This proof is divided into three steps. In (a) we first demonstrate that Assumption 3.2 holds given Assumption 4.3. In (b) we derive continuity of our stage potential function $\boldsymbol{p}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$ according to (3.42) and lastly in (c) we establish an upper bound on our rotated stage potential function $\tilde{\boldsymbol{p}}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$ as defined in (3.43).

(a) To establish uniform strict convexity of the stage potential function $\boldsymbol{p}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$, we examine the convexity condition (3.25) with $\theta \in [0, 1]$ and the combined state $\boldsymbol{z} := (\boldsymbol{x}_t, \boldsymbol{u}_t)$ where

$\boldsymbol{z}_1, \boldsymbol{z}_2 \in \boldsymbol{\mathcal{X}} \times \boldsymbol{\mathcal{U}}$. We further define the stacked matrices and vectors

$$\boldsymbol{M}_t := \text{blkdiag}(\boldsymbol{Q}_t, \boldsymbol{R}_t), \quad \boldsymbol{m}_t := \begin{bmatrix} \boldsymbol{s}_t & \boldsymbol{c}_t \end{bmatrix} \tag{4.25}$$

allowing us to compactly express the convexity condition (3.25) as follows

$$\begin{aligned}
\boldsymbol{p}_t(\theta \boldsymbol{z}_1 + (1-\theta)\boldsymbol{z}_2) &\leqslant \theta \boldsymbol{p}_t(\boldsymbol{z}_1) + (1-\theta)\boldsymbol{p}_t(\boldsymbol{z}_2) - \kappa\,\theta(1-\theta)\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) \\
\kappa\,\theta(1-\theta)\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) &\leqslant \theta \boldsymbol{p}_t(\boldsymbol{z}_1) + (1-\theta)\boldsymbol{p}_t(\boldsymbol{z}_2) - \boldsymbol{p}_t(\theta \boldsymbol{z}_1 + (1-\theta)\boldsymbol{z}_2) \\
\kappa\,\theta(1-\theta)\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) &\leqslant \theta \boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_1 + (1-\theta)\boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 - (\theta \boldsymbol{z}_1 + (1-\theta)\boldsymbol{z}_2)^\top \boldsymbol{M}_t (\theta \boldsymbol{z}_1 + (1-\theta)\boldsymbol{z}_2)
\end{aligned} \tag{4.26}$$

where $\kappa\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|)$ denotes the lower bound on the convexity condition that we seek to find. Note that the linear cost terms $\boldsymbol{m}_t$ cancel out in the convexity condition, as they are affine and therefore satisfy the condition exactly. Through the following series of algebraic manipulations, we rewrite this expression to obtain the sought lower bound

$$\begin{aligned}
\kappa\,\theta(1-\theta)\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) &\leqslant \theta \boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_1 + (1-\theta)\boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 - (\theta(\boldsymbol{z}_1 - \boldsymbol{z}_2) + \boldsymbol{z}_2)^\top \boldsymbol{M}_t (\theta(\boldsymbol{z}_1 - \boldsymbol{z}_2) + \boldsymbol{z}_2) \\
\kappa\,\theta(1-\theta)\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) &\leqslant \theta \boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_1 + (1-\theta)\boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 - \theta^2(\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t (\boldsymbol{z}_1 - \boldsymbol{z}_2) \\
&\qquad - 2\theta(\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t \boldsymbol{z}_2 - \boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 \\
\kappa\,\theta(1-\theta)\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) &\leqslant \theta \boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_1 + (1-\theta)\boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 - \theta^2(\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t (\boldsymbol{z}_1 - \boldsymbol{z}_2) - 2\theta \boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_2 \\
&\qquad + 2\theta \boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 - \boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 \\
\kappa\,\theta(1-\theta)\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) &\leqslant \theta \boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_1 - 2\theta \boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_2 + \theta \boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 - \theta^2(\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t (\boldsymbol{z}_1 - \boldsymbol{z}_2) \\
\kappa\,\theta(1-\theta)\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) &\leqslant \theta(\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t (\boldsymbol{z}_1 - \boldsymbol{z}_2) - \theta^2(\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t (\boldsymbol{z}_1 - \boldsymbol{z}_2) \\
\kappa\,\theta(1-\theta)\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) &\leqslant \theta(1-\theta)(\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t (\boldsymbol{z}_1 - \boldsymbol{z}_2) \\
\kappa\,\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) &\leqslant (\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t (\boldsymbol{z}_1 - \boldsymbol{z}_2)
\end{aligned} \tag{4.27}$$

giving us uniform strict convexity with $\kappa := 1$ and $\gamma(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) := \underline{\boldsymbol{\sigma}}_z \|\boldsymbol{z}_1 - \boldsymbol{z}_2\|$ where $\underline{\boldsymbol{\sigma}}_z := \min(\underline{\boldsymbol{\sigma}}_x, \underline{\boldsymbol{\sigma}}_u)$.

(b) To show continuity we start with Definition (3.42) from Assumption 3.10

$$\begin{aligned}
|\boldsymbol{p}_t(\boldsymbol{z}_1) - \boldsymbol{p}_t(\boldsymbol{z}_2)| &= |\boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_1 + \boldsymbol{m}_t \boldsymbol{z}_1 - \boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 - \boldsymbol{m}_t \boldsymbol{z}_2| && (4.28) \\
&= |\boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_1 - \boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_2 + \boldsymbol{z}_1^\top \boldsymbol{M}_t \boldsymbol{z}_2 - \boldsymbol{z}_2^\top \boldsymbol{M}_t \boldsymbol{z}_2 + \boldsymbol{m}_t(\boldsymbol{z}_1 - \boldsymbol{z}_2)| && (4.29) \\
&= |\boldsymbol{z}_1^\top \boldsymbol{M}_t(\boldsymbol{z}_1 - \boldsymbol{z}_2) + (\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t \boldsymbol{z}_2 + \boldsymbol{m}_t(\boldsymbol{z}_1 - \boldsymbol{z}_2)| && (4.30) \\
&\overset{\text{(I)}}{\leqslant} |\boldsymbol{z}_1^\top \boldsymbol{M}_t(\boldsymbol{z}_1 - \boldsymbol{z}_2)| + |(\boldsymbol{z}_1 - \boldsymbol{z}_2)^\top \boldsymbol{M}_t \boldsymbol{z}_2| + |\boldsymbol{m}_t(\boldsymbol{z}_1 - \boldsymbol{z}_2)| && (4.31) \\
&\overset{\text{(II)}}{\leqslant} \|\boldsymbol{M}_t\| \big(\|\boldsymbol{z}_1\| + \|\boldsymbol{z}_2\|\big) \cdot \|\boldsymbol{z}_1 - \boldsymbol{z}_2\| + \|\boldsymbol{m}_t\| \cdot \|\boldsymbol{z}_1 - \boldsymbol{z}_2\| && (4.32)
\end{aligned}$$

where we first apply the triangle inequality in (I), followed by the induced matrix norm combined with the Cauchy-Schwarz inequality in (II). An upper bound for this expression is given by

$$\gamma_{\boldsymbol{p}}(\|\boldsymbol{z}_1 - \boldsymbol{z}_2\|) := (2\,\overline{\boldsymbol{\sigma}}_z \overline{\boldsymbol{z}} + \overline{\boldsymbol{m}})\|\boldsymbol{z}_1 - \boldsymbol{z}_2\| \tag{4.33}$$

where $\overline{\boldsymbol{\sigma}}_z := \max(\overline{\boldsymbol{\sigma}}_x, \overline{\boldsymbol{\sigma}}_u)$, $\overline{\boldsymbol{z}} := \max_{\boldsymbol{z} \in \boldsymbol{\mathcal{Y}}} \|\boldsymbol{z}\|$ and $\overline{\boldsymbol{m}} := \max(\overline{\boldsymbol{s}}, \overline{\boldsymbol{c}})$. The existence of $\overline{\boldsymbol{z}}$ is guaranteed by the compactness of $\boldsymbol{\mathcal{Y}}$, which is a subset of $\boldsymbol{\mathcal{X}} \times \boldsymbol{\mathcal{U}}$ as specified in Assumption 3.10.

(c) We are left to find an upper bound on our rotated potential function $\tilde{\boldsymbol{p}}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$ which we derive from it's Definition 3.9

$$\begin{aligned}
\tilde{\boldsymbol{p}}_t(\boldsymbol{z}_t) &= \hat{\boldsymbol{p}}_t(\boldsymbol{z}_t) + \lambda_t(\boldsymbol{x}_t) - \lambda_{t+1}(\boldsymbol{A}_t \boldsymbol{x}_t + \boldsymbol{B}_t \boldsymbol{u}_t + \boldsymbol{d}_t) && (4.34) \\
&\overset{(3.27)}{=} \boldsymbol{z}_t^\top \boldsymbol{M}_t \boldsymbol{z}_t - (\boldsymbol{z}_t^\star)^\top \boldsymbol{M}_t \boldsymbol{z}_t^\star + \boldsymbol{m}_t(\boldsymbol{z}_t - \boldsymbol{z}_t^\star) + \underbrace{\frac{1}{\eta_0}\begin{bmatrix} q_{t+1}^\top \boldsymbol{A}_t - q_t^\top & q_{t+1}^\top \boldsymbol{B}_t \end{bmatrix}(\boldsymbol{z}_t - \boldsymbol{z}_t^\star)}_{\lambda_t(\boldsymbol{x}_t) - \lambda_{t+1}(\boldsymbol{A}_t \boldsymbol{x}_t + \boldsymbol{B}_t \boldsymbol{u}_t + \boldsymbol{d}_t)} && (4.35) \\
&= \boldsymbol{z}_t^\top \boldsymbol{M}_t \boldsymbol{z}_t - (\boldsymbol{z}_t^\star)^\top \boldsymbol{M}_t \boldsymbol{z}_t^\star + \tilde{\boldsymbol{m}}_t(\boldsymbol{z}_t - \boldsymbol{z}_t^\star) && (4.36)
\end{aligned}$$

where we substituted the storage function $\lambda_t(\boldsymbol{x}_t)$ (3.27) provided by Theorem 3.5. Further we introduce the combined vector $\tilde{\boldsymbol{m}}_t := \boldsymbol{m}_t + \begin{bmatrix} q_{t+1}^\top \boldsymbol{A}_t - q_t^\top & q_{t+1}^\top \boldsymbol{B}_t \end{bmatrix} / \eta_0$ to keep the notation lean. Given expression can be upper bounded as follows

$$\begin{aligned}
\tilde{\boldsymbol{p}}_t(\boldsymbol{z}_t) &= (\boldsymbol{z}_t - \boldsymbol{z}_t^\star)^\top \boldsymbol{M}_t (\boldsymbol{z}_t - \boldsymbol{z}_t^\star) + 2(\boldsymbol{z}_t^\star)^\top \boldsymbol{M}_t \boldsymbol{z}_t - 2(\boldsymbol{z}_t^\star)^\top \boldsymbol{M}_t \boldsymbol{z}_t^\star + \tilde{\boldsymbol{m}}_t(\boldsymbol{z}_t - \boldsymbol{z}_t^\star) && (4.37)\\
&= (\boldsymbol{z}_t - \boldsymbol{z}_t^\star)^\top \boldsymbol{M}_t (\boldsymbol{z}_t - \boldsymbol{z}_t^\star) + 2(\boldsymbol{z}_t^\star)^\top \boldsymbol{M}_t (\boldsymbol{z}_t - \boldsymbol{z}_t^\star) + \tilde{\boldsymbol{m}}_t(\boldsymbol{z}_t - \boldsymbol{z}_t^\star) && (4.38)\\
&\overset{\text{(I)}}{\leqslant} \|(\boldsymbol{z}_t - \boldsymbol{z}_t^\star)^\top \boldsymbol{M}_t (\boldsymbol{z}_t - \boldsymbol{z}_t^\star)\| + \|2(\boldsymbol{z}_t^\star)^\top \boldsymbol{M}_t (\boldsymbol{z}_t - \boldsymbol{z}_t^\star)\| + \|\tilde{\boldsymbol{m}}_t(\boldsymbol{z}_t - \boldsymbol{z}_t^\star)\| && (4.39)\\
&\overset{\text{(II)}}{\leqslant} \|\boldsymbol{M}_t\| \cdot \|\boldsymbol{z}_t - \boldsymbol{z}_t^\star\|^2 + \big(2\|\boldsymbol{M}_t\|\|\boldsymbol{z}_t^\star\| + \|\tilde{\boldsymbol{m}}_t\|\big) \cdot \|\boldsymbol{z}_t - \boldsymbol{z}_t^\star\| && (4.40)
\end{aligned}$$

where we first upper-bounded the expression with its norm in (I) and than applied the matrix norm together with the Cauchy-Schwarz inequality similar to (4.32). We obtain the final upper bound required by Assumption 3.10 given by

$$\gamma_{\tilde{\boldsymbol{p}}}(\|\boldsymbol{z}_t - \boldsymbol{z}_t^\star\|) := \overline{\boldsymbol{\sigma}}_z \|\boldsymbol{z}_t - \boldsymbol{z}_t^\star\|^2 + (2\,\overline{\boldsymbol{\sigma}}_z \overline{\boldsymbol{z}}^\star + \overline{\tilde{\boldsymbol{m}}})\|\boldsymbol{z}_t - \boldsymbol{z}_t^\star\| \tag{4.41}$$

where $\overline{\boldsymbol{\sigma}}_z := \max(\overline{\boldsymbol{\sigma}}_x, \overline{\boldsymbol{\sigma}}_u)$, $\overline{\boldsymbol{z}}^\star := \max_{t \in \mathbb{N}} \|\boldsymbol{z}_t^\star\|$ and $\overline{\tilde{\boldsymbol{m}}} := \max_{t \in \mathbb{N}} \|\tilde{\boldsymbol{m}}_t\|$. The upper bound on the infinite-horizon optimal trajectory $\overline{\boldsymbol{z}}^\star$ exists due to Assumption 4.3 which requires a nonzero cost on state and input which will inevitably lead to a bounded state and input sequence if the problem is feasible. The combined vector $\|\tilde{\boldsymbol{m}}_t\|$ is bounded since it writes as

$$\begin{aligned}
\|\tilde{\boldsymbol{m}}_t\| &= \Big\|\boldsymbol{m}_t + \frac{1}{\eta_0}\begin{bmatrix} q_{t+1}^\top \boldsymbol{A}_t - q_t^\top & q_{t+1}^\top \boldsymbol{B}_t \end{bmatrix}\Big\| && (4.42)\\
&\leqslant \|\boldsymbol{m}_t\| + \frac{1}{\eta_0}\Big\|\begin{bmatrix} q_{t+1}^\top \boldsymbol{A}_t - q_t^\top & q_{t+1}^\top \boldsymbol{B}_t \end{bmatrix}\Big\| && (4.43)\\
&\leqslant \overline{\boldsymbol{\sigma}}_z + \frac{1}{\eta_0}\Big\|\begin{bmatrix} q_{t+1}^\top \boldsymbol{A}_t - q_t^\top & q_{t+1}^\top \boldsymbol{B}_t \end{bmatrix}\Big\| && (4.44)
\end{aligned}$$

with $\overline{\boldsymbol{\sigma}}_z := \max(\overline{\boldsymbol{\sigma}}_x, \overline{\boldsymbol{\sigma}}_u)$ provided by Assumption 4.3 and the bounded Lagrange multipliers η_0 and q_t from Theorem 3.3[1].

□

This brings us to one of our main results. With Assumptions 4.3, 4.4, and 3.9 in place, we can invoke Lemma 4.5 which ensures that all the assumptions from [13] hold, such that we can apply [13, Theorem 1,2,4] as outlined in Section 3.2.2 providing us with $\mathbb{P}$-practical asymptotic stability. We numerically verify these results in a simulation study presented in Chapter 5.

4.4 Stability with Terminal Ingredients

In this section, we establish asymptotic stability for the centralized OCP

$$\begin{aligned}
V_N(\boldsymbol{x}(t), t) := \min_{\boldsymbol{x}_{\cdot|t}, \boldsymbol{u}_{\cdot|t}} \quad & \boldsymbol{p}_N\big(\boldsymbol{x}_{\cdot|t}, \boldsymbol{u}_{\cdot|t}, t\big)\\
\text{s.t.} \quad & \boldsymbol{x}_{0|t} = \boldsymbol{x}(t),\\
& \boldsymbol{x}_{k+1|t} = \boldsymbol{f}_{t+k}(\boldsymbol{x}_{k|t}, \boldsymbol{u}_{k|t}), \quad \forall\, k \in \mathbb{N}_N\\
& (\boldsymbol{x}_{k|t}, \boldsymbol{u}_{k|t}) \in \boldsymbol{\mathcal{X}} \times \boldsymbol{\mathcal{U}}, \qquad \forall\, k \in \mathbb{N}_N\\
& \boldsymbol{x}_{N|t} = \boldsymbol{x}_{t+N}^r,
\end{aligned} \tag{4.45}$$

using the time-varying reference trajectory $\boldsymbol{x}^r$ as an exact terminal constraint and the potential function $\boldsymbol{p}_N(\boldsymbol{x}_t, \boldsymbol{u}_t, t)$ (4.14) as the cost, leveraging the theory outlined in Section 3.2.3. Specifically,

[1] The boundedness of the Lagrange multipliers is established in the proof of Theorem 3.3 and is essential for ensuring the theorem's validity.

we consider the LTV dynamics defined in (4.22), along with the compact, convex state and input constraint set defined by

$$(\boldsymbol{x}_t, \boldsymbol{u}_t) \in \boldsymbol{\mathcal{X}} \times \boldsymbol{\mathcal{U}} := \big\{(\boldsymbol{x}, \boldsymbol{u}) \in \mathbb{R}^{n+m} \mid \boldsymbol{h}(\boldsymbol{x}_t, \boldsymbol{u}_t) \leqslant 0\big\}. \tag{4.46}$$

where $\boldsymbol{h}(\boldsymbol{x}_t, \boldsymbol{u}_t) := \big(\boldsymbol{h}_x(\boldsymbol{x}_t), \boldsymbol{h}_u(\boldsymbol{u}_t)\big) \in \mathbb{R}^c$ is a continuous and differentiable function that stacks the state and input constraints and hence does not introduce any coupling between state and inputs.

Additionally, we assume that our system is periodically time-varying as stated in the assumption below. This assumption is needed to enable us to compute a dissipative reference trajectory, which is exceedingly hard for general time-varying systems.

Assumption 4.6. [Periodic Problem] The dynamics and cost are periodically time-varying with the known period $T \in \mathbb{N}$ according to

$$\boldsymbol{f}_t(\boldsymbol{x}_t, \boldsymbol{u}_t) = \boldsymbol{f}_{t+T}(\boldsymbol{x}_t, \boldsymbol{u}_t), \quad \boldsymbol{p}_t(\boldsymbol{x}_t, \boldsymbol{u}_t) = \boldsymbol{p}_{t+T}(\boldsymbol{x}_t, \boldsymbol{u}_t). \tag{4.47}$$

Remark 13. The analysis presented herein also accounts for the more general case where individual agents v have systems with different periods T^v by defining T as the least common multiple of $T^v \, \forall \, v \in \mathcal{A}$.

The following assumption ensures boundedness and positive definiteness of the cost allowing potentially indefinite matrices in the stage potential function $\boldsymbol{p}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$. It can also be interpreted as a stage-vise convexity condition.

Assumption 4.7. [Assumptions on Cost] There exist periodically time-varying, symmetric matrices $\boldsymbol{K}_t \in \mathbb{R}^{n \times n}$ with $\boldsymbol{K}_t = \boldsymbol{K}_{t+T}$ such that the following linear matrix inequality (LMI)

$$\underline{\boldsymbol{\sigma}}_z \mathcal{I} \preceq \begin{bmatrix} \boldsymbol{Q}_t + \boldsymbol{K}_t - \boldsymbol{A}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{A}_t & -\boldsymbol{A}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{B}_t \\ -\boldsymbol{B}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{A}_t & \boldsymbol{R}_t - \boldsymbol{B}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{B}_t \end{bmatrix} \preceq \overline{\boldsymbol{\sigma}}_z \mathcal{I} \tag{4.48}$$

holds with the constants $\underline{\boldsymbol{\sigma}}_z, \overline{\boldsymbol{\sigma}}_z > 0$ for all $t \in \mathbb{N}_T$ effectively rendering the problem uniformly strictly convex (see Definition 3.2). We further assume that the linear cost terms are bounded

$$\|\boldsymbol{s}_t\| \leqslant \overline{\boldsymbol{s}} \quad \|\boldsymbol{c}_t\| \leqslant \overline{\boldsymbol{c}} \tag{4.49}$$

with the constants $\overline{\boldsymbol{s}}, \overline{\boldsymbol{c}} \geqslant 0$ for all $t \in \mathbb{N}_T$.

In [51] the same inequality as (4.48) is studied for linear quadratic costs in a time-invariant setup with an LTI dynamics. Therein they establish global dissipativity and Turnpike with respect to a steady-state by analyzing the above LMI, relating it to system properties such as observability and the systems eigenvalues.

Remark 14. The above inequality (4.48) is trivially fulfilled for all $t \in \mathbb{N}_T$ with $\boldsymbol{Q}_t$ and $\boldsymbol{R}_t$ being positive definite as defined in (4.23) and $\boldsymbol{K}_t = 0$.

The next assumption is needed to guarantee strong duality of problem (4.51) [65, 5.2.3 Strong duality and Slater's constraint qualification].

Assumption 4.8. [Slater's Condition] There exists at least one strictly feasible trajectory $\boldsymbol{x}$ such that

$$\big\{\boldsymbol{x} \mid \boldsymbol{x}_{k+1} = \boldsymbol{f}_k(\boldsymbol{x}_k, \boldsymbol{u}_k), \; \boldsymbol{x}_T = \boldsymbol{x}(t), \; \boldsymbol{h}(\boldsymbol{x}_k, \boldsymbol{u}_k) < 0, \quad \forall \, k \in \mathbb{N}_T\big\} \neq \varnothing \tag{4.50}$$

is nonempty.

Under the given assumptions, a dissipative reference trajectory $\boldsymbol{x}^r$ can be obtained according to the following theorem, using any dual-method (cf. CVXOPT, MOSEK, Gurobi, OSQP).

Theorem 4.9. *Suppose Assumptions 4.6 and 4.8 hold. Then, a dissipative reference trajectory* $(\boldsymbol{x}^r, \boldsymbol{u}^r)$ *for problem* (4.45) *can be obtained as the solution to the following OCP*

$$\begin{aligned} (\boldsymbol{x}^r, \boldsymbol{u}^r) := \underset{\boldsymbol{x},\boldsymbol{u}}{\arg\min} \quad & \boldsymbol{p}_T(\boldsymbol{x}, \boldsymbol{u}) \\ s.t. \quad & \boldsymbol{x}_{k+1} = \boldsymbol{f}_k(\boldsymbol{x}_k, \boldsymbol{u}_k), \quad \forall\, k \in \mathbb{N}_T \\ & (\boldsymbol{x}_k, \boldsymbol{u}_k) \in \boldsymbol{\mathcal{X}} \times \boldsymbol{\mathcal{U}}, \quad \forall\, k \in \mathbb{N}_T \\ & \boldsymbol{x}_T = \boldsymbol{x}(t) \end{aligned} \tag{4.51}$$

rendering the system strictly dissipative for all $t \in \mathbb{N}$ *on the set* $\boldsymbol{\mathcal{X}} \times \boldsymbol{\mathcal{U}}$*, with respect to the supply rate* $\boldsymbol{s}_t(\boldsymbol{x}_t, \boldsymbol{u}_t) := \hat{\boldsymbol{p}}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$ *and the reference trajectory* $(\boldsymbol{x}^r, \boldsymbol{u}^r)$*. The corresponding storage function is given by*

$$\lambda_t(\boldsymbol{x}_t) = -q_{t \bmod T}^\top (\boldsymbol{x}_t - \boldsymbol{x}^r_{t \bmod T}). \tag{4.52}$$

where $q_t \in \mathbb{R}^n$ *are the Lagrange multipliers of problem* (4.51)*.*

Proof. In this proof, we will establish strict dissipativity with respect to the reference trajectory $\boldsymbol{x}^r$ computed for our periodical problem (4.51). First, we will show uniform strict convexity of the potentially positive indefinite stage potential function $\boldsymbol{p}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$ by rotating it using a suited periodically time-varying function $\boldsymbol{r}(\boldsymbol{x}_t) \in \mathbb{R}$

$$\boldsymbol{r}(\boldsymbol{x}_t) := \boldsymbol{x}_t^\top \boldsymbol{K}_t \boldsymbol{x}_t, \qquad \boldsymbol{r}_t(\boldsymbol{x}_t) = \boldsymbol{r}_{t+T}(\boldsymbol{x}_t) \tag{4.53}$$

without changing the solution to our OCPs (4.45) and (4.51). Second we will demonstrate strict dissipativity similar to the proof of Theorem 3.5 with the difference that we show this with state and input constraints for finite horizons.

The intermediate stage cost $\boldsymbol{\varphi}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$ is obtained by rotating the stage potential function $\boldsymbol{p}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)$ according to

$$\boldsymbol{\varphi}_t(\boldsymbol{x}_t, \boldsymbol{u}_t) := \boldsymbol{p}_t(\boldsymbol{x}_t, \boldsymbol{u}_t) + \boldsymbol{r}_t(\boldsymbol{x}_t) - \boldsymbol{r}_{t+1}(\boldsymbol{f}_t(\boldsymbol{x}_t, \boldsymbol{u}_t)). \tag{4.54}$$

Unlike in (3.28), the purpose of this rotation is not to establish strict dissipativity, but to show that a potentially indefinite stage potential function can, in principle, be transformed into a uniformly strictly convex function by applying a suitable rotation. By plugging in (4.53) and the dynamics defined in (4.22) we can write

$$\boldsymbol{\varphi}_t(\boldsymbol{x}_t, \boldsymbol{u}_t) = \boldsymbol{p}_t(\boldsymbol{x}_t, \boldsymbol{u}_t) + \boldsymbol{x}_t^\top (\boldsymbol{K}_t - \boldsymbol{A}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{A}_t) \boldsymbol{x}_t - 2\boldsymbol{x}_t^\top \boldsymbol{A}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{B}_t \boldsymbol{u}_t - \boldsymbol{u}_t^\top \boldsymbol{B}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{B}_t \boldsymbol{u}_t \tag{4.55}$$

$$\overset{(4.16)}{=} \begin{bmatrix} \boldsymbol{x} & \boldsymbol{u} \end{bmatrix} \begin{bmatrix} \boldsymbol{Q}_t + \boldsymbol{K}_t - \boldsymbol{A}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{A}_t & -\boldsymbol{A}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{B}_t \\ -\boldsymbol{B}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{A}_t & \boldsymbol{R}_t - \boldsymbol{B}_t^\top \boldsymbol{K}_{t+1} \boldsymbol{B}_t \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{u} \end{bmatrix} + \boldsymbol{s}_t \boldsymbol{x}_t + \boldsymbol{c}_t \boldsymbol{u}_t \tag{4.56}$$

which is uniformly strictly convex given that Assumption 4.7 holds (cf. proof of Lemma 4.5). We provide a numerical examples in (4.1) where we successfully convexify an indefinite stage potential function by rotation. The respective cost of the intermediate stage cost is given by

$$\boldsymbol{\Phi}_{T|t} = \sum_{k=t}^{t+T-1} \boldsymbol{\varphi}_k(\boldsymbol{x}_k, \boldsymbol{u}_k) = \boldsymbol{p}_N(\boldsymbol{x}, \boldsymbol{u}, t) + \boldsymbol{r}_t(\boldsymbol{x}_t) - \boldsymbol{r}_{t+T}(\boldsymbol{x}_{t+T}) \tag{4.57}$$

which does not change the solution to OCP (4.45) as both initial and terminal state are fixed. Similarly, the solution to OCP (4.51) remains unaffected, since both the rotation function $\boldsymbol{r}_t(\boldsymbol{x}_t) = \boldsymbol{r}_{t+T}(\boldsymbol{x}_t)$ and the trajectory $\boldsymbol{x}_t = \boldsymbol{x}_{t+T}$ are periodic with period T and hence cancel each other out.

To derive strict dissipativity we begin by stating the dual problem of OCP (4.51) and by analyzing the respective KKT conditions and its implications. More specifically we will find an update rule for the multipliers that allows us to demonstrate strict dissipativity analogously to the proof of Theorem 3.5.

The dual problem of OCP (4.51) is given by

$$
\begin{aligned}
(\boldsymbol{x}^r, \boldsymbol{u}^r, q, \mu) := \underset{\boldsymbol{x},\boldsymbol{u},q,\mu}{\arg\min} \quad & \boldsymbol{L}(\boldsymbol{x}, \boldsymbol{u}, q, \mu) \\
\text{s.t.} \quad & \mu_k \geqslant 0, \quad \forall\, k \in \mathbb{N}_T
\end{aligned}
\tag{4.58}
$$

with the Lagrangian defined by

$$
\boldsymbol{L}(\boldsymbol{x}, \boldsymbol{u}, q, \mu) := q_0^\top(\boldsymbol{x}_T - \boldsymbol{x}_0) + \sum_{k=0}^{T-1} \boldsymbol{\varphi}_t(\boldsymbol{x}_t, \boldsymbol{u}_t) + q_{t+1}^\top(\boldsymbol{f}_t(\boldsymbol{x}_t, \boldsymbol{u}_t) - \boldsymbol{x}_{t+1}) + \mu_t^\top \boldsymbol{h}(\boldsymbol{x}_t, \boldsymbol{u}_t) \tag{4.59}
$$

and its Lagrange multipliers $q_0, q_{t+1} \in \mathbb{R}^n$ and $\mu_t \in \mathbb{R}^c$ for all $t \in \mathbb{N}_T$. Under Assumptions 4.7 and 4.8, and together with the convexity of the constraint set, the KKT conditions presented below are both necessary and sufficient to guarantee primal and dual optimality (cf. [65, Chapter 5.5.3]).

$$
\begin{aligned}
\text{Primal Feasibility} \left\{ \right. & \begin{aligned} \boldsymbol{x}_T^r &= \boldsymbol{x}^r(t), & \\ \boldsymbol{x}_{k+1}^r &= \boldsymbol{f}_k(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r), & \forall\, k \in \mathbb{N}_T \\ \boldsymbol{h}(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r) &\leqslant 0, & \forall\, k \in \mathbb{N}_T \end{aligned} \\
\text{Complementary Slackness} \left\{ \right. & \mu_k^\top \boldsymbol{h}(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r) = 0, & \forall\, k \in \mathbb{N}_T \\
\text{Dual Feasibility} \left\{ \right. & \mu_k \geqslant 0, & \forall\, k \in \mathbb{N}_T \\
\text{Stationarity} \left\{ \right. & \begin{aligned} \left.\frac{\partial \boldsymbol{L}}{\partial \boldsymbol{x}_k}\right|_{(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r)} &= 0, & \forall\, k \in \mathbb{N}_{T+1} \\ \left.\frac{\partial \boldsymbol{L}}{\partial \boldsymbol{u}_k}\right|_{(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r)} &= 0. & \forall\, k \in \mathbb{N}_T \end{aligned}
\end{aligned}
\tag{4.60}
$$

Examining the stationarity condition reveals the following periodic relationship for the multipliers

$$
q_0 = q_T \tag{4.61}
$$

$$
q_k^\top = q_{k+1}^\top \left.\frac{\partial \boldsymbol{f}_k}{\partial \boldsymbol{x}_k}\right|_{(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r)} + \left.\frac{\partial \boldsymbol{\varphi}_k}{\partial \boldsymbol{x}_k}\right|_{(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r)} + \mu_k^\top \left.\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}_k}\right|_{(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r)} \quad \forall\, k \in \mathbb{N}_T \tag{4.62}
$$

$$
0 = q_{k+1}^\top \left.\frac{\partial \boldsymbol{f}_k}{\partial \boldsymbol{u}_k}\right|_{(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r)} + \left.\frac{\partial \boldsymbol{\varphi}_k}{\partial \boldsymbol{u}_k}\right|_{(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r)} + \mu_k^\top \left.\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{u}_k}\right|_{(\boldsymbol{x}_k^r, \boldsymbol{u}_k^r)} \quad \forall\, k \in \mathbb{N}_T \tag{4.63}
$$

which enables us to establish strict dissipativity using the storage function defined in (4.52). The remainder of the proof follows analogously to the proof of Theorem 3.5, with $\eta_0 = 1$ being normalized[2] and the constraints $\boldsymbol{h}(\boldsymbol{x}_t, \boldsymbol{u}_t)$, where the optimal steady-state trajectory $(\boldsymbol{x}^\star, \boldsymbol{u}^\star)$ is replaced by the periodically varying optimal reference trajectory $(\boldsymbol{x}^r, \boldsymbol{u}^r)$. □

Remark 15. For the special case of a period of $T = 1$, i.e., a time-invariant problem, we recover the results from [51, Chapter 4], where global strict dissipativity is established under Assumption 4.7.

Given the above results we can now state our second main result. Under the stated conditions and the structure of the cost and dynamics, Assumptions 3.12-3.15 are satisfied. Asymptotic stability follows from Theorem 3.17, provided the optimal rotated value function $\tilde{V}_N(\boldsymbol{x}(t), t)$ is continuous, as required by Assumption 3.16.

Remark 16. Continuity of the optimal rotated value function $\tilde{V}_N(\boldsymbol{x}(t), t)$ as stated by Assumption 3.16, can be established similar to Section 3.2.2 under the assumption of dissipativity, a suitable controllability condition together with continuity and boundedness of the stage cost as described in [49]. We can achieve that by introducing Assumptions 3.9 and 3.7, which, together with Assumption 4.7 and Lemma 4.5, allow us to apply [49, Theorem 2] as performed in Section 3.2.2, Lemma 3.11.

[2] Since the set of all multipliers forms a cone, we can normalize them, allowing us to choose $\eta_0 = 1$ (cf. [61, Equation 2.3]).

Example 4.1 *Rotation of Indefinite Cost.* Let's consider the scalar 2-periodic LTV system

$$x_{t+1} = a_t x_t + b_t u_t, \qquad a_0 = 2,\ a_1 = -2,\ b_0 = 2,\ b_1 = 4 \tag{4.64}$$

together with the periodic quadratic cost defined below

$$\ell_t(x_t, u_t) = q_t x_t^2 + r_t u_t^2, \qquad q_0 = 2,\ q_1 = 1,\ r_0 = 0,\ r_1 = -1. \tag{4.65}$$

The system remains unstable at all times, while the input cost r_t transitions from being positive semi-definite to indefinite. However, the cost can be rendered positive definite by applying a suitable rotation with $k_0 = k_1 = -\frac{1}{2}$, using the rotation function defined in (4.53). The corresponding LMIs for this rotation are

$$(t = 0) : \begin{bmatrix} 2.5 & 2 \\ 2 & 2 \end{bmatrix}, \quad (t = 1) : \begin{bmatrix} 2.5 & 4 \\ 4 & 7 \end{bmatrix} \tag{4.66}$$

derived from (4.48). Both of these LMIs are positive definite, thereby rendering the system strongly convex. It is important to note that no solution exists for $|b_1| < 2$ or when $b_0 = 0$. In general, the conditions under which this rotation is possible remain unclear and is subject for further investigation. It is assumed that they relate to the statements of [69, Theorem 1] where they analyze stability with an indefinite state cost for a p-step system utilizing results from [51]. Figure 4.1 depicts the inequalities that render the LMI (4.48) positive definite with the intersection being the set of possible solutions for k_0, k_1. A derivation of these inequalities can be found in Appendix A.

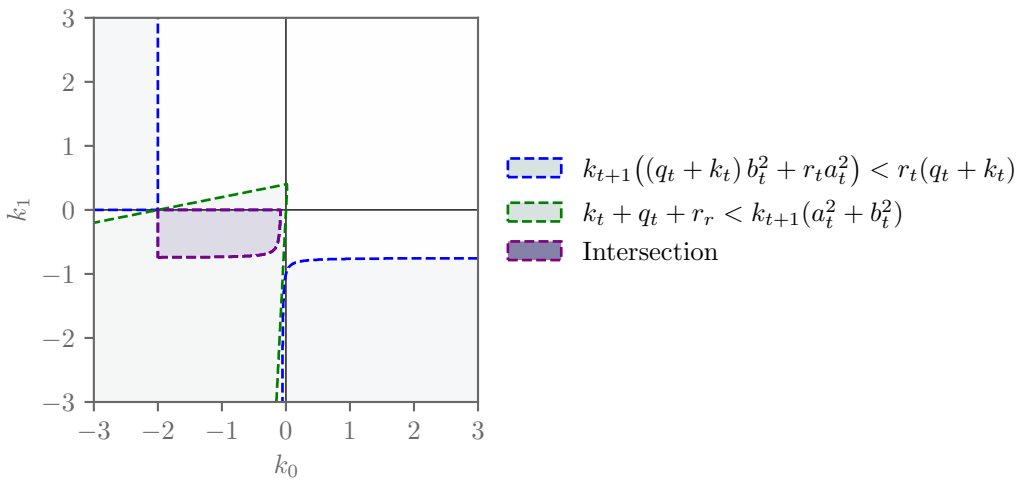

Figure 4.1: Inequalities that render LMI (4.48) positive definite, with the linear inequalities (green shaded) and nonlinear inequalities (blue shaded). The purple shaded area represents their intersection, i.e., all possible solutions for k_0 and k_1.

In the following chapter, we examine the theoretical results derived in this chapter and compare the methods discussed here with those from Section 4.3.

Chapter 5

Simulation Studies

In this Chapter we numerically verify and compare the two methods proposed in the previous Chapter 4 on an example from demand-side management.

5.1 Model: Demand-Side Management

The example considered is taken from [7] and is motivated by the challenge of maintaining grid stability in privatized energy systems, where prosumers, i.e., individuals who both consume and produce energy, cannot be directly controlled. In such systems, prosumers must be incentivized to support grid stability, for instance, by adjusting electricity prices, while operational and aggregate load limits are enforced to ensure safe grid operation. This approach is commonly referred to as DSM. In this context, our focus will be on demonstrating asymptotic stability and properties involved such as the Turnpike phenomenon. We will compare both previously proposed methods against each other in terms of performance, stability and feasibility.

We begin by translating the problem into our notation and by laying out the dynamics, constraints and cost. The dynamics of the individual agents [7, Equation 9] is given with the following system matrices

$$A_t^v := \begin{bmatrix} 1 & 0 \\ 0 & \alpha_t^v \end{bmatrix}, \quad B_t^v := \begin{bmatrix} 1 & 0 \\ 0 & \beta_t^v \end{bmatrix}, \quad d_t^v := \begin{bmatrix} -{e_t}^{v,\mathrm{ref}} \\ 0 \end{bmatrix} \tag{5.1}$$

with α_t^v and β_t^v being the leakage rate and the charging efficiency, respectively and ${e_t}^{v,\mathrm{ref}}$ being the nominal consumption. For each prosumer we define the state vector $x_t^v := (\zeta_t^v, q_t^v)$ which collects the energy debt and the battery state of charge (SoC) and the input vector $u_t^v := (e_t^v, s_t^v)$ which collects the energy consumption and the battery charging/discharging inputs.

The constraints are defined as follows

$$\zeta_t^v \in \left[\underline{\zeta}_t^v, \quad \overline{\zeta}_t^v\right], \qquad q_t^v \in \left[0, \quad \overline{q}^v\right] \tag{5.2}$$

$$e_t^v \in \left[\underline{e}^v, \quad \overline{e}^v\right], \qquad s_t^v \in \left[\underline{s}^v, \quad \overline{s}^v\right] \tag{5.3}$$

$$e_t^v + s_t^v \in \left[g_t^v, g_t^v + \overline{l}^v\right], \qquad \boldsymbol{L}_t \in \left[\underline{\boldsymbol{L}}_t^v, \overline{\boldsymbol{L}}_t^v\right] \tag{5.4}$$

where the aggregative load $\boldsymbol{L}_t$ introduces a coupling constraint among the agents, as it is defined as the aggregate of their inputs, given by

$$\boldsymbol{L}_t := \sum_{v \in \mathcal{A}} e_t^v + s_t^v - g_t^v + \sum_{v \in \mathcal{P}} e_t^v. \tag{5.5}$$

Here, $\mathcal{P}$ denotes the set of passive consumers that do not participate in the DSM scheme.

Remark 17. The constraints defined above violate two assumption. First, we have time-varying constraints, which are not currently considered due to the additional complexity they introduce. However, it may be possible to adapt the theory to accommodate time-varying constraints, as both [13], which underpins the findings in Section 3.2.2, and [70], which supports the results in Section 3.2.3, are valid for general time-varying constraints. Second, we include state constraints, which are currently not supported by the theory of section 3.2.2 due to the dependence on the strong maximum principle 3.3. This principle, as of now, only permits input constraints (see Remark 6).

In a next step we recast the stage cost defined in [7, Equation 10] into matrix notation.

Proposition 5.1. *The stage cost presented in [7, Equation 10] can be rewritten in matrix from as defined in* (4.11) *with the matrices*

$$Q_t^{i,i} = \begin{bmatrix} \gamma_{1,t}^i & 0 \\ 0 & \gamma_{2,t}^i \end{bmatrix} \quad \text{and} \quad R_t^{i,j} = \begin{bmatrix} \rho_{1,t}^i & \rho_{1,t}^i \\ \rho_{1,t}^i & \rho_{1,t}^i \end{bmatrix} \qquad \forall\, i,j \in \mathcal{A}. \tag{5.6}$$

and the vectors $s_t^v = 0$, $c_t^v = \left(\rho_{1,t}^v c_{1,t} - \rho_{1,t}^v\, g_t^v + \rho_{2,t}^v\right)\begin{bmatrix}1 & 1\end{bmatrix}$, *with* $c_{1,t}$ *being the uncontrolled aggregate load defined by*

$$c_{1,t} := -\sum_{v\in\mathcal{A}_g} g_t^v + \sum_{v\in\mathcal{P}} e_t^v \tag{5.7}$$

with $\mathcal{A}_g$ *denoting the subset of prosumers that additionally produce* g_t^v *units of non-dispatchable power by sources such as solar energy. Furthermore, the positive constants* $\rho_{1,t}^v$ *and* $\rho_{2,t}^v$ *represent the different price rates that prosumers have previously negotiated with suppliers. Meanwhile,* $\gamma_{1,t}^v$ *and* $\gamma_{2,t}^v$ *are positive weights penalizing the energy shift and battery usage, respectively.*

Proof. Let's start by restating the stage cost of agent v as defined in [7, Equation 10]

$$\ell_t^v(x_t^v, u_t^v, \boldsymbol{u}_t^{-v}) = \rho_{1,t}^v L_t l_t^v + \rho_{2,t}^v l_t^v + \|\zeta_t^v\|^2_{\gamma_{1,t}^v} + \|q_t^v\|^2_{\gamma_{2,t}^v} \tag{5.8}$$

and the aggregate load as introduced in [7, Equation 2]

$$\begin{aligned} L_t &= \sum_{v\in\mathcal{A}} l_t^v + \sum_{v\in\mathcal{P}} l_t^v = \sum_{v\in\mathcal{A}} \left(e_t^v + s_t^v\right) - \sum_{v\in\mathcal{A}_g} g_t^v + \sum_{v\in\mathcal{P}} e_t^v && (5.9)\\ &= \sum_{v\in\mathcal{A}} \left(e_t^v + s_t^v\right) + c_{1,t} && (5.10)\\ &= \sum_{v\in\mathcal{A}} \left(\begin{bmatrix}1 & 1\end{bmatrix} u_t^v\right) + c_{1,t} && (5.11) \end{aligned}$$

with $c_{1,t}$ defined according to (5.7). Furthermore, we have l_t^v defined as in [7, Equation 1]

$$l_t^v = e_t^v + s_t^v - g_t^v = \begin{bmatrix}1 & 1\end{bmatrix} u_t^v - g_t^v. \tag{5.12}$$

and by plugging (5.12) and (5.11) into (5.8) we write

$$\begin{aligned} \ell_t^v(x_t^v, u_t^v, \boldsymbol{u}_t^{-v}) = \rho_{1,t}^v \underbrace{\left(\left(u_t^v\right)^\top \begin{bmatrix}1\\1\end{bmatrix} - g_t^v\right)}_{l_t^v} \underbrace{\left(\sum_{v\in\mathcal{A}} \left(\begin{bmatrix}1 & 1\end{bmatrix} u_t^v\right) + c_{1,t}^v\right)}_{L_t^v} \\ + \rho_{2,t}^v \underbrace{\left(\begin{bmatrix}1 & 1\end{bmatrix} u_t^v - g_t^v\right)}_{l_t^v} + \left(x_t^v\right)^\top Q_t^{v,v}\, x_t^v. \end{aligned} \tag{5.13}$$

with the state cost matrix $Q_t^{v,v}$ introduced in (5.6). After expanding this expression we get

$$\ell_t^v(x_t^v, u_t^v, \boldsymbol{u}_t^{-v}) = \rho_{1,t}^v \sum_{i\in\mathcal{A}} \left((u_t^v)^\top \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} u_t^i \right) + (\rho_{1,t}^v c_1 + \rho_{2,t}^v) \begin{bmatrix} 1 & 1 \end{bmatrix} u_t^v$$
$$- (\rho_{1,t}^v c_1 + \rho_{2,t}^v)\, g_t^v - \rho_{1,t}^v g_t^v \sum_{v\in\mathcal{A}} \begin{bmatrix} 1 & 1 \end{bmatrix} u_t^v + (x_t^v)^\top Q_t^{v,v}\, x_t^v \quad (5.14)$$

which can further be rewritten to

$$\ell_t^v(x_t^v, u_t^v, \boldsymbol{u}_t^{-v}) = \sum_{i\in\mathcal{A}} \left((u_t^v)^\top R_t^{v,i}\, u_t^i \right) + (\rho_{1,t}^v c_1 + \rho_{2,t}^v + \rho_{1,t}^v g_t^v) \begin{bmatrix} 1 & 1 \end{bmatrix} u_t^v + (x_t^v)^\top Q_t^{v,v}\, x_t^v$$
$$- (\rho_{1,t}^v c_{1,t}^v + \rho_{2,t}^v) - \rho_{1,t}^v g_t^v \sum_{i\in\mathcal{A}\setminus v} \begin{bmatrix} 1 & 1 \end{bmatrix} u_t^i \quad (5.15)$$
$$= (x_t^v)^\top Q_t^{v,v}\, x_t^v + \sum_{i\in\mathcal{A}} \left((u_t^v)^\top R_t^{v,i}\, u_t^i \right) + c_t^v u_t^v + c_{2,t}^v \quad (5.16)$$

with c_t^v as defined in (5.6) and $c_{2,t}^v = -(\rho_{1,t}^v c_{1,t}^v + \rho_{2,t}^v) - \rho_{1,t}^v g_t^v \sum_{i\in\mathcal{A}\setminus v} [1 \quad 1]\, u_t^i$ being a constant independent of agent v's decision variables. □

Remark 18. From Equation (5.6) one can conclude that the input cost matrix $\boldsymbol{R}_t$ is only positive-semi definite which violates Assumption 4.3. In simulation it was found that we can rotate this semi-definite cost to be positive definite according to the method described in 3.16.

In the following, we present the simulation results obtained using `Python` and CVXOPT [71, 72]. The parameters are the same as those in [7], except that we simulate the system with a periodicity of 96 hours (4 days) instead of 48 hours. This adjustment helps mitigate boundary effects and better approximate the behavior of a non-periodically time-varying system. In other words, we extend the simulation horizon to emulate a continuously time-varying system more accurately. Furthermore we approximate the infinite-horizon optimal solution $(\boldsymbol{x}^\star, \boldsymbol{u}^\star)$ to our system by choosing a horizon of 576 hours which corresponds to 24 days.

5.2 EMPC without Terminal Ingredients

In Figure 5.1 we can see multiple closed-loop trajectories of agent 10 starting from different initial conditions, converging to the infinite-horizon optimal trajectory $(\boldsymbol{x}^\star, \boldsymbol{u}^\star)$. One can observe that all of the trajectories have converged to the approximate infinite-horizon optimal solution within 36 hours with a minimum deviation visible in the inputs. A similar trend can be observed for all agents. One trajectory was also initialized on the state constraint boundary directly, which turned out to be infeasible.

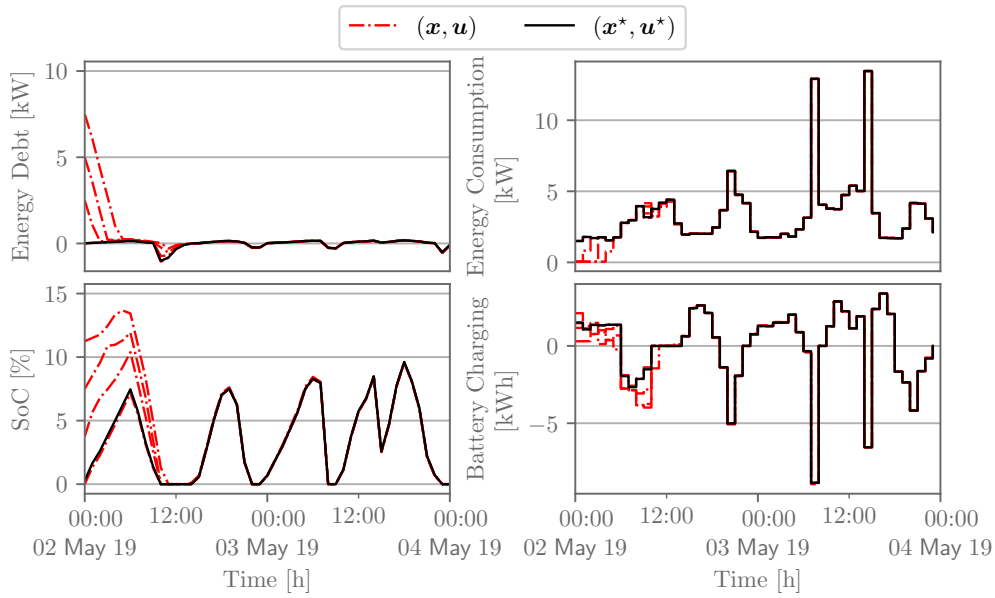

Figure 5.1: Closed-loop trajectories of agent 10 (x^{10}, u^{10}) (red) asymptotically converging to the approximate optimal overtaking trajectory $(\boldsymbol{x}^\star, \boldsymbol{u}^\star)$ (black) starting from different initial conditions with a horizon $N = 24$.

In Figure 5.2, we show the L2 distance between the closed-loop trajectories, initialized from different starting conditions, and the approximate infinite-horizon optimal trajectory. As before, we observe that all trajectories converge to a neighborhood around the optimal trajectory. Interestingly, the size of this neighborhood does not appear to shrink as the horizon increases, contrary to what we would expect based on [13, Theorem 4]. Additionally, note that the trajectory starting from $x_0^v = [10.0,\ 15.0]$ is infeasible.

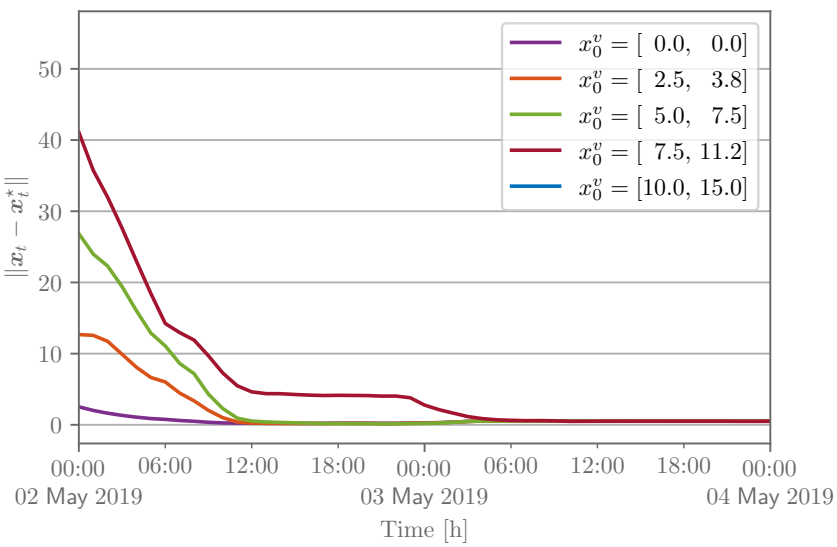

Figure 5.2: Asymptotic convergence of the L2 distance between the closed-loop trajectory $\boldsymbol{x}$ and the the approximate optimal overtaking trajectory $x^\star$ starting from different initial conditions with a horizon $N = 24$.

Figure 5.3 illustrates the so-called Turnpike property. To demonstrate this, we simulated trajectories from the same initial condition with horizon lengths ranging from 6 to 48 hours. As shown, all open-loop trajectories remain close to the approximate infinite-horizon optimal trajectory before eventually deviating. This behavior suggests that the system is dissipative with respect to the infinite-horizon optimal trajectory.

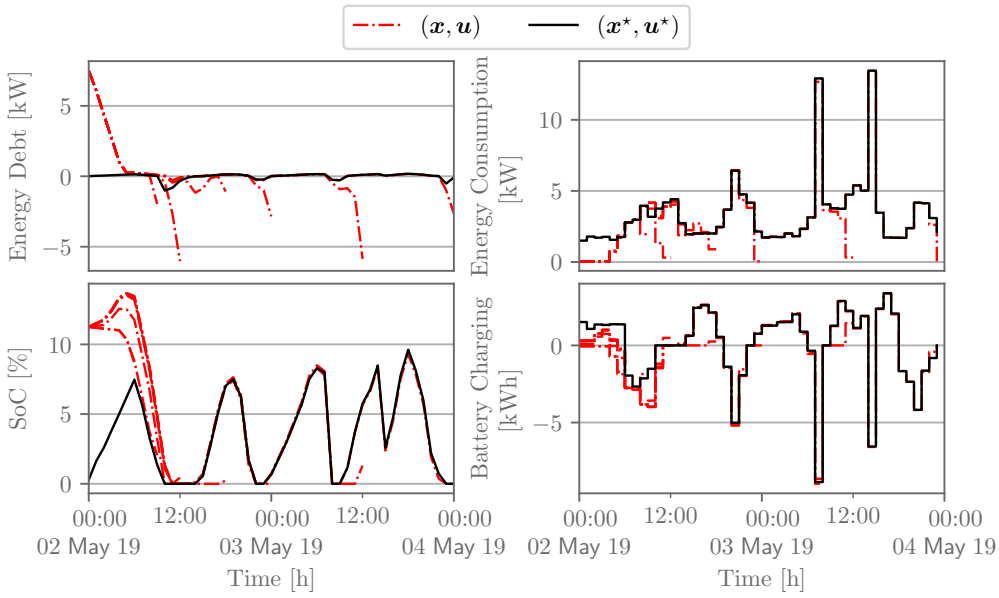

Figure 5.3: Open-loop predictions $(x^{10}_{\bullet|t}, u^{10}_{\bullet|t})$ (red) of agent 10 at $t = 0$ for increasing horizon lengths N, demonstrating turnpike with respect to approximate optimal overtaking trajectory $\boldsymbol{x}^\star$ (black). The trajectories where initialized at $x^v_0 = [7.5,\ 11.25]$.

A similar observation can be made from Figure 5.4, with the key difference that shorter-horizon trajectories, such as those with horizons of 6 and 9 hours, diverge much earlier from the others. While this is not inherently problematic, it suggests that horizons of this length may no longer guarantee proper convergence to the infinite-horizon optimal trajectory.

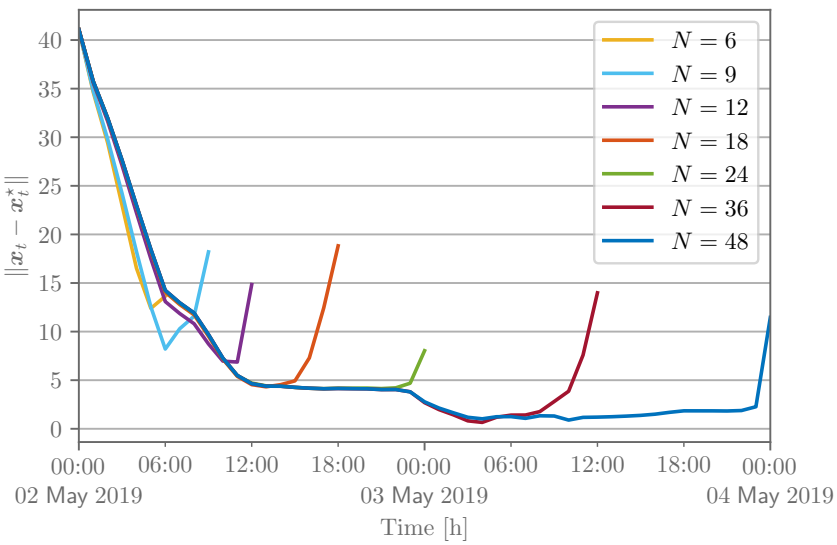

Figure 5.4: Turnpike being apparent for the L2 distance between the open-loop predictions $\boldsymbol{x}_{\bullet|t}$ and the approximate optimal overtaking trajectory $\boldsymbol{x}^\star$ at $t = 0$ for increasing horizon lengths N. The trajectories where initialized at $x^v_0 = [7.5,\ 11.25]$.

This concluding Figure 5.5 combines Figure 5.4 and 5.2 by showing closed-loop predictions together with open-loop predictions for a horizon of 6 hours. One can see that our previous concerns are confirmed as the closed-loop trajectory does not effectively converge to the approximate overtaking trajectory anymore.

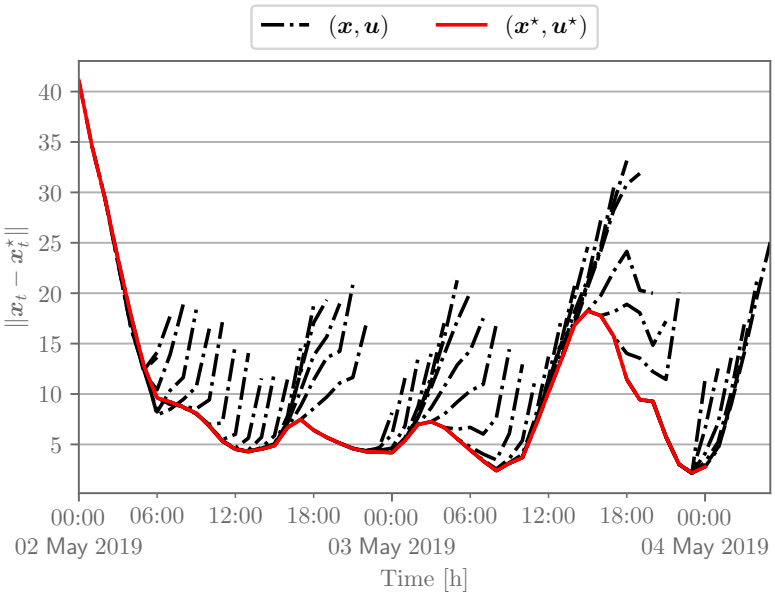

Figure 5.5: L2 distance between the open-loop predictions $\boldsymbol{x}_{\cdot|t}$, starting on their respective closed-loop trajectory $\boldsymbol{x}$ (red), and the approximate optimal overtaking trajectory $\boldsymbol{x}^\star$ (black) for horizon $N = 6$.

5.3 EMPC with Periodic Reference Trajectory

Figure 5.6 compares the Lagrange multipliers obtained from solving the approximate infinite-horizon problem with those from solving OCP (4.51) with a period $T = 96$, alongside the propagated multipliers based on the multiplier dynamics in (4.61), as predicted by Theorem 4.9. It is not surprising that the multiplier trajectories align closely, as they all effectively approximate the same periodic OCP, in accordance with Assumption 4.6.

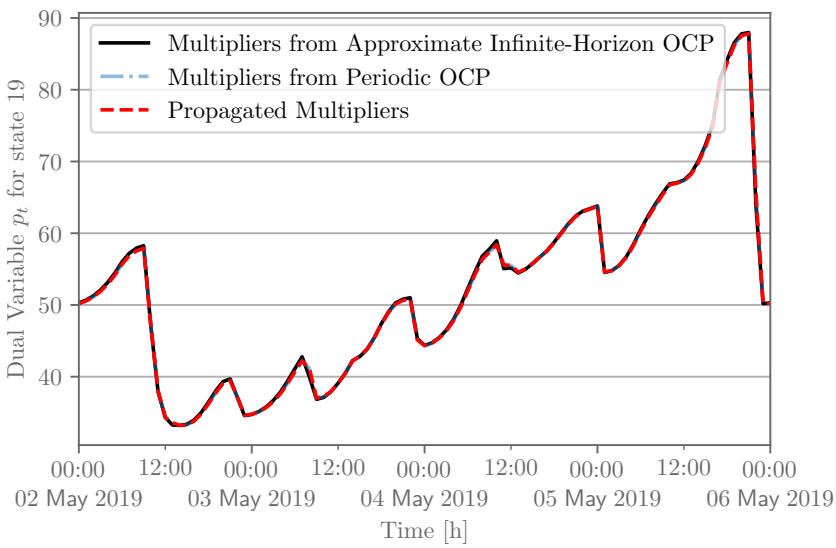

Figure 5.6: Lagrange multipliers of state 19 for the periodic solution of OCP (4.51) and the approximate infinite-horizon solution showing periodicity according to the dynamics predicted in (4.62).

Figure 5.7 depicts the L2 distance between the periodic reference trajectory and the closed-loop trajectories initialized from different starting conditions. Two key observations can be made. First, infeasibility occurs earlier compared to the OCP without terminal constraints, which is expected since terminal constraints inherently reduce the region of attraction. Second, the closed-loop trajectories exhibit better convergence to the dissipative reference trajectory than those shown in Figure 5.2. This behavior is somewhat surprising, and we do not yet have a complete explanation for it.

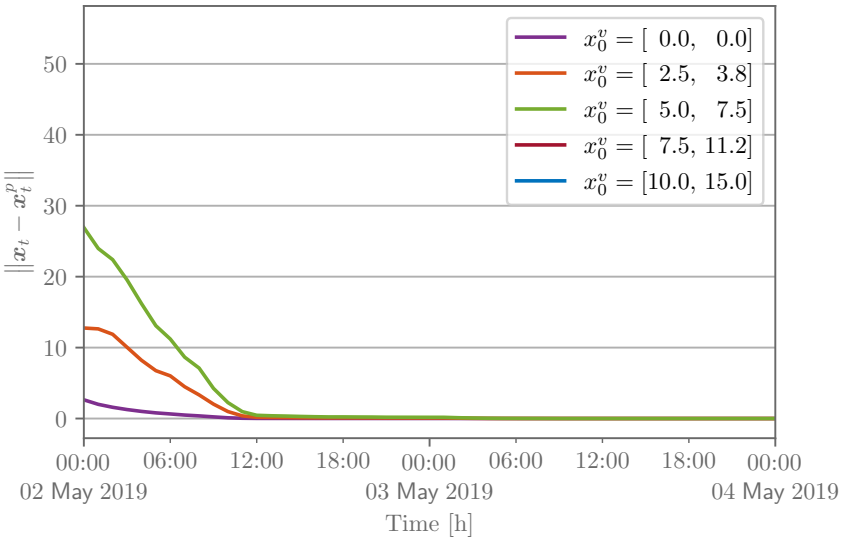

Figure 5.7: Asymptotic convergence of the L2 distance between the closed-loop trajectory $\boldsymbol{x}$ and the the approximate optimal overtaking trajectory $x^\star$ starting from different initial conditions with a horizon $N = 24$.

Figure 5.8 illustrates the feasible region of the EMPC with terminal constraints for different horizon lengths. Notably, a horizon of at least 36 hours is required to achieve a feasible set comparable in size to that of the EMPC without terminal constraints.

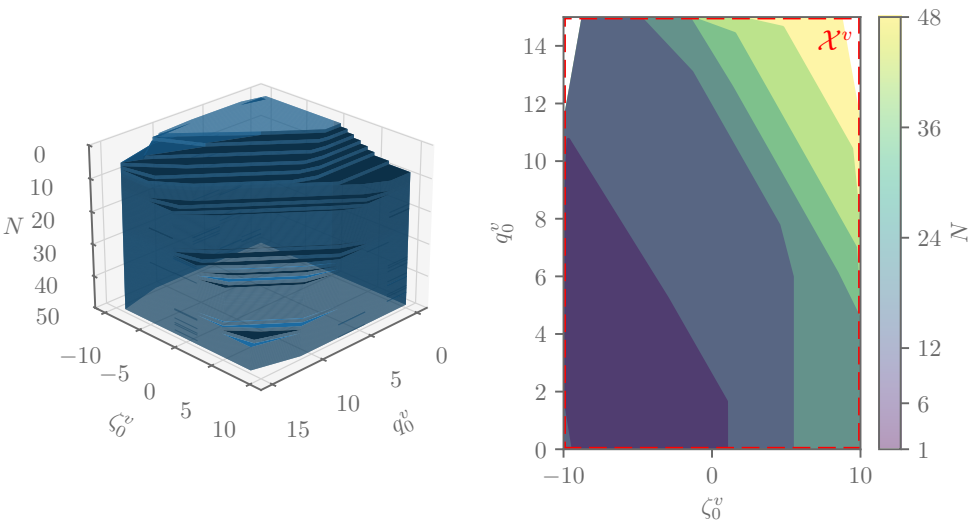

Figure 5.8: Feasible region for different horizon lengths N given that each agent $v \in \mathcal{A}$ starts from the same initial state $x_0^v = (\zeta_0^v, q_0^v)$ at $t = 0$. The state constraints are shown as a red dashed box.

The table below compares the closed-loop performance of the two control schemes over a 48-hours with a horizon of 24 hours. The results show a cost deviation of only 6 ‰, indicating that both schemes perform equally well in terms of closed-loop performance.

Initial State x_0^v	Closed-Loop Cost		Relative Difference
[kW, %]	Without	With	[‰]
[0.0, 0.0]	30299	30492	6
[2.5, 3.8]	27918	28110	6
[5.0, 7.5]	27541	27731	6
[7.5,11.3]	30191	-	-

Table 5.1: Closed-loop cost comparison with and without terminal constraint for a horizon of $N = 24$. The dash denotes infeasible solutions.

Chapter 6

Conclusion

In this work, we derived stability and recursive feasibility guarantees for receding horizon games under a potential game structure with a linear quadratic objective and linear time-varying dynamics. To this end, we first provided an overview on potential games and economic model predictive control theory, laying the foundation for a clear connection between the two topics. We then established the assumptions under which the considered RHG attains a PG structure, enabling it to be reformulated as a centralized OCP which can be analyzed using tools from EMPC theory.

Building on this connection, we first derived asymptotic stability and recursive feasibility for the game without additional terminal ingredients, showing convergence to the infinite-horizon optimal trajectory. We achieved that for LTV systems with input constraints, by imposing suitable controllability assumptions (i.e., local controllability 3.9 and n_x-step reachability 4.4) for bounded, positive definite potential functions (see Assumption 4.3). The main strength of this approach lies in its simplicity, as it provides stability certificates without additional terminal ingredients, such as terminal costs or constraints. However, this simplicity comes at the cost of strong assumptions which may fail to hold even for simple LTI systems, such as local controllability[1](3.9).

A second approach was studied where we analyzed stability and recursive feasibility for RHGs with periodically time-varying LTV systems using a dissipative reference trajectory as an exact terminal constraint. We achieved this for general state and input constraints by employing an LMI condition on the cost (4.7), allowing it to be potentially indefinite, combined with an assumption on the rotated optimal value function (3.16). This LMI condition introduces a time-varying rotation of the stage potential function, enabling a potentially indefinite linear quadratic game to become positive definite. Furthermore, we demonstrate how to compute a dissipative reference trajectory for this periodically time-varying LTV system, while also identifying a corresponding storage function. The key advantage of this approach is its ability to guarantee recursive feasibility, unlike the method with terminal constraints, which remains feasible only if the horizon is sufficiently long. Additionally, even when dissipativity of the reference trajectory is lost, this approach provides an asymptotic average performance bound, allowing for an estimation of the game's performance. However, a significant drawback of this method is the reduced feasible region caused by the exact terminal constraint. Furthermore, it can be impractical for certain applications to adapt the game to satisfy this additional terminal constraint.

Finally, we verified the two proposed methods in a simulation study based on an example from demand-side management. In simulation both of the approaches, EMPC with and without terminal constraint, showed convergence to the dissipative reference trajectory and the infinite-horizon optimal trajectory, respectively. Additionally, the simulations demonstrated that both methods can

[1]This situation can arise, for example, when the optimal trajectory lies at the boundary of the constraints. Depending on the dynamics, it might then not be possible anymore to control the system around the δ_c-neighborhood of the optimal trajectory.

handle general state and input constraints, which is currently not supported by the theory for the approach with no terminal ingredients. A comparison of the two methods further revealed an almost identical closed-loop cost when simulated over a 48-hour period and showed great potential for real world applications. Notably, both methods successfully handled a positive semi-definite input cost, a feature that is not yet supported by the theory of the method without terminal ingredients.

6.1 Outlook

Although these methods worked fine in simulation, both approaches are still limited by a number of different factors.

The main issue for the method without terminal constraints is that we are currently not able to consider any state constraints due to the lack of respective infinite-horizon theorems. The reasons for this are discussed in Remark 6. Further work could include extending the infinite-horizon theorem 3.3 to general state and input constraints under reasonable assumptions. Additionally, overcoming the current limitation to LTV systems with convex cost functions remains an open challenge, requiring further exploration of different approaches to establish dissipativity for potentially nonlinear systems with non-convex cost functions.

The approach with terminal ingredients is limited by similar issues. Here we also depend on LTV dynamics with the additional assumption of periodicity. It would be interesting to explore alternative, non-dual based, methods to compute dissipative reference trajectories for non-periodic and even nonlinear dynamics. Another subject for further investigation is under which condition the LMI condition proposed in Assumption 4.7 can be satisfied. Also, we require very similar conditions for the two approaches, even though intuitively one would expect to have less strong assumptions for the method with terminal constraints. Hence, relaxing the conditions of the approach with terminal constraints is a promising research direction.

Lastly, it would be interesting to analyze other RHG with varying cost functions, couplings, and dynamics. Additionally, extending the approach to non-potential games, such as the one discussed in [4], could be a promising direction for future research.

Appendix A

Auxiliary Results

A.1 Proof of Example 4.1

In order to derive the inequalities related to the cost rotation of the scalar LTV system from Example 4.1, we start by analyzing the respective LMI condition 4.48 which writes as

$$\underline{\sigma}_z \mathcal{I} \preceq \underbrace{\begin{bmatrix} q_t + k_t - k_{t+1}a_t^2 & -k_{t+1}a_t b_t \\ -k_{t+1}a_t b_t & r_t - k_{t+1}b_t^2 \end{bmatrix}}_{M_t} \preceq \overline{\sigma}_z \mathcal{I} \tag{A.1}$$

with the positive constant $\underline{\sigma}_z, \overline{\sigma}_z > 0$. This condition is satisfied if the matrix M_t has only positive eigenvalues λ_t. Hence, we can find a set of inequalities by computing the eigenvalues λ_t of M_t. The eigenvalues are computed as follows

$$\begin{aligned}
\mathrm{eig}(M_t) \Rightarrow 0 &= \det(M_t - \lambda_t \mathcal{I}_2) \\
&= \det \begin{bmatrix} q_t + k_t - k_{t+1}a_t^2 - \lambda_t & -k_{t+1}a_t b_t \\ -k_{t+1}a_t b_t & r_t - k_{t+1}b_t^2 - \lambda_t \end{bmatrix} \\
&= (q_t + k_t - k_{t+1}a_t^2 - \lambda_t)(r_t - k_{t+1}b_t^2 - \lambda_t) - k_{t+1}^2 a_t^2 b_t^2 \\
&= \lambda_t^2 - \lambda_t(r_t - k_{t+1}b_t^2 + q_t + k_t - k_{t+1}a_t^2) + (q_t + k_t - k_{t+1}a_t^2)(r_t - k_{t+1}b_t^2) \\
&\qquad - k_{t+1}^2 a_t^2 b_t^2 \\
&= \lambda_t^2 + \lambda_t\big(k_{t+1}(a_t^2 + b_t^2) - q_t - r_t - k_t\big) + (q_t + k_t)(r_t - k_{t+1}b_t^2) - k_{t+1}a_t^2 r_t \\
&\qquad + k_{t+1}a_t^2 b_t^2 - k_{t+1}^2 a_t^2 b_t^2 \\
&= \lambda_t^2 + \lambda_t\big(k_{t+1}(a_t^2 + b_t^2) - q_t - r_t - k_t\big) - \big(k_{t+1}\big((q_t + k_t)b_t^2 + a_t^2 r_t\big) - r_t(q_t + k_t)\big)
\end{aligned} \tag{A.2}$$

which leaves us with a quadratic equation. We can solve this quadratic equation by applying the quadratic formula which gives us

$$\begin{aligned}
\lambda_t = &-\frac{1}{2}\big(k_{t+1}(a_t^2 + b_t^2) - q_t - r_t - k_t\big) \\
&\pm \frac{1}{2}\sqrt{\underbrace{\big(k_{t+1}(a_t^2 + b_t^2) - q_t - r_t - k_t\big)^2}_{c_1} + \underbrace{4\big(k_{t+1}\big((q_t + k_t)b_t^2 + a_t^2 r_t\big) - r_t(q_t + k_t)\big)}_{c_2}}.
\end{aligned} \tag{A.3}$$

where we additionally define the intermediate variables c_1 and c_2 for later. From this expression we obtain our first inequality which is given by

$$-\big(k_{t+1}(a_t^2 + b_t^2) - q_t - r_t - k_t\big) > 0 \tag{A.4}$$

$$q_t + r_t + k_t > k_{t+1}(a_t^2 + b_t^2). \tag{A.5}$$

This expression could also have been derived from the fact that trace$(M_t) > 0$, as it represents the sum of the eigenvalues of M_t. The following inequality holds for nonnegative values under the square root, $c_1 + c_2$, a condition we will verify in a later step. To ensure that λ_t remains positive for all times we establish the following inequality

$$-\big(k_{t+1}(a_t^2+b_t^2)-q_t-r_t-k_t\big) > \sqrt{\big(k_{t+1}(a_t^2+b_t^2)-q_t-r_t-k_t\big)^2+4\big(k_{t+1}\big((q_t+k_t)b_t^2+a_t^2r_t\big)-r_t(q_t+k_t)\big)}$$

$$0 > 4\big(k_{t+1}\big((q_t+k_t)b_t^2+a_t^2r_t\big)-r_t(q_t+k_t)\big) \tag{A.6}$$

$$r_t(q_t+k_t) > k_{t+1}\big((q_t+k_t)b_t^2+a_t^2r_t\big) \tag{A.7}$$

which provides the second inequality for the cost rotation of Example 4.1. We are left to show that the expression below the square root $c_1 + c_2$ will always take nonnegative values. In order to do that we first expand the first expression c_1

$$\begin{aligned} c_1 = \big(k_{t+1}(a_t^2+b_t^2)-q_t-r_t-k_t\big)^2 &= k_{t+1}^2(a_t^2+b_t^2)^2 - 2k_{t+1}a_t^2q_t - 2k_{t+1}b_t^2q_t - 2k_{t+1}a_t^2r_t - 2k_{t+1}b_t^2r_t \\ &\quad - 2k_{t+1}k_ta_t^2 - 2k_{t+1}k_tb_t^2 + 2q_tr_t + 2k_tq_t + 2k_tr_t + q_t^2 + r_t^2 + k_t^2 \end{aligned} \tag{A.8}$$

and we now expand the second expression c_2

$$c_2 = 4\big(k_{t+1}\big((q_t+k_t)b_t^2+a_t^2r_t\big)-r_t(q_t+k_t)\big) = 4k_{t+1}b_t^2q_t + 4k_{t+1}k_tb_t^2 + 4k_{t+1}a_t^2r_t - 4q_tr_t - 4k_tr_t. \tag{A.9}$$

By combining Expression (A.8) and (A.9) we write

$$\begin{aligned} c_1 + c_2 &= k_{t+1}^2(a_t^2+b_t^2)^2 - 2k_{t+1}a_t^2q_t + 2k_{t+1}b_t^2q_t + 2k_{t+1}a_t^2r_t - 2k_{t+1}b_t^2r_t \\ &\quad - 2k_{t+1}k_ta_t^2 + 2k_{t+1}k_tb_t^2 - 2q_tr_t + 2k_tq_t - 2k_tr_t + q_t^2 + r_t^2 + k_t^2 && \text{(A.10)} \\ &= k_{t+1}^2(a_t^2+b_t^2)^2 - 2k_{t+1}(a_t^2-b_t^2)(r_t-q_t-k_t) + (r_t-q_t-k_t)^2 && \text{(A.11)} \\ &\overset{\text{(I)}}{\leqslant} k_{t+1}^2(a_t^2-b_t^2)^2 - 2k_{t+1}(a_t^2-b_t^2)(r_t-q_t-k_t) + (r_t-q_t-k_t)^2 && \text{(A.12)} \\ &\leqslant \big(k_{t+1}(a_t^2-b_t^2) - (r_t-q_t-k_t)\big)^2 && \text{(A.13)} \end{aligned}$$

where we used the lower bound $(a^2 - b^2) \leqslant (a^2 + b^2)$ in (I). Since Expression A.13 is quadratic it will always remain nonnegative and hence Inequality (A.5) and (A.7) need to hold for all times.

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